

The Impact of New Products on Ethical Beliefs*

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Abstract

Does technological change affect ethical beliefs? We study the case of synthetic alternatives to meat products using an incentivized survey experiment. One might expect that the possibility of an improved alternative would increase objections to meat production on moral grounds because it reduces the cost of avoiding meat. Instead, we find that informing consumers of a new alternative decreases moral objections to conventional animal farming relative to a control. Participants who are experimentally nudged to view the new alternative relatively more positively do indeed report more moral concern for animal welfare. The findings suggest a backlash effect in which people react to innovation by increasing their support for existing practices, but depending on how positively they view the alternative.

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1. Introduction

When new technologies spread, former practices sometimes come to appear morally repugnant. For example, attitudes toward child labor and smog in urban areas changed in Western countries simultaneously with increased mechanization and advancements in combustion technology, respectively (Edmonds 2008, Reitze 1999). A simple story could explain this pattern: When a transaction raises potential ethical concerns, consumers may resolve this dilemma by downplaying the ethical implications of their choice (Hestermann et al. 2020, Bénabou and Tirole 2016, Rabin 1994). The more appealing the transaction, the greater the cost to consumers of entertaining objections to it. The invention of an alternative makes it easier to make a different choice, which can in turn lead to increased willingness to consider concerns with the established way of doing things.

We examine whether information about a new alternative, “cultured” meat (i.e., lab-grown, *in vitro*, clean, or cultivated meat), causes individuals to change their moral beliefs about animals and meat products.¹ Cultured meat is grown from animal cell cultures, yielding a product that is chemically identical to conventional meat but does not require animals to be raised and slaughtered. It is currently undergoing taste tests or special releases in small quantities. In principle, those who believe that consuming meat is necessary for health could find this product a closer substitute to traditional meat products than the existing plant-based substitutes on the market. Some consumers may also expect cultured meat to be a closer substitute in terms of taste. Since the production of cultured meat does not involve the slaughter of animals, it could make traditional meat production appear more morally repugnant.

To test this hypothesis, we randomly assign 6,000 individuals in the U.S. who are part of a nationally representative panel to alternatively read about a conventional farm meat,

¹In our main questionnaire, we use the term “clean meat.” As companies developing synthetic meat products have grown in the past several years, the common term for the product has changed. At the time of survey design, clean meat was the common industry term, though now “cultivated” and “cultured” are somewhat more common. A separate experiment used the term “cultured meat” and obtained similar results. We use the term “cultured” meat going forward.

plant-based meat, or cultured meat product after answering a baseline questionnaire. Our experiment communicates that this is a new product not yet on the market, and we use a realistic description developed in coordination with a prominent cultured meat company.² So as to have an exogenous instrument of the desirability of the substitute, those assigned to learn about cultured meat are randomly selected to read negative comments from other respondents about the new product and read either an article defending it or a placebo article, while those assigned to the farm meat or plant-based meat treatment arms view the placebo article. We then show respondents a video of an undercover investigation into a large pig farm that supplies Walmart. If information about the new product makes participants more open to considering objections to farm meat production, this video could shift their views. We then measure respondents’ moral concerns for animals, willingness to donate to a farmed-animal protection organization, relative willingness-to-pay for farm meat and plant-based meat in an actual choice, and, for those informed about cultured meat products, relative willingness-to-pay for farm meat and cultured meat.

Our experiments provide a unique opportunity to isolate the effects of awareness and interest in a real emerging technology on moral attitudes and economic behavior. If consuming farm meat motivates consumers to avoid moral concerns with conventional animal farming, as in Hestermann et al. (2020), then the existence of a realistic alternative might increase consumers’ willingness to entertain moral concern with conventional animal farming or place more moral weight on animals.³ Relatedly, it might decrease willingness to pay for farm meat relative to a plant-based alternative. The more attractive the alternative, the more we might expect it to displace farm meat and make participants more likely to entertain moral concerns for animals. In line with this last intuition, our main, pre-registered, analysis finds that those we experimentally nudge to have more positive attitudes toward an alternative have more moral concerns with conventional farm meat production as measured

²While coordinating with a cultured meat company was essential for developing realistic experimental materials, their sharing of material on cultured meat was the extent of their involvement and they did not have the right to review our results.

³In Appendix A, we present a theoretical model that generalizes Hestermann et al. (2020) to identify conditions under which a new meat alternative increases moral concerns with animal farming.

by both stated beliefs and willingness to pay. However, those shown information about the alternative on its own exhibit reduced moral concerns relative to those shown information about a conventional farm meat product. In other words, two things are simultaneously true: those experimentally exposed to cultured meat do not express greater moral concern for animals than those exposed to farm meat, but those experimentally nudged to view the cultured meat product relatively more favorably do show more concern for animals than those who were not nudged in this way. We discuss several possible explanations for these results, most notably that they would be consistent with a potential backlash effect or what is called “reactance” in the psychology literature (Brehm and Brehm 2013).

This study contributes to a growing body of work that considers the economics of animal welfare. Hestermann et al. (2020) develop a motivated reasoning model for meat consumption, but do not consider how the introduction of new alternatives might affect this reasoning. In perhaps the most closely related paper, Espinosa and Treich (2023) conduct a lab experiment with French students in which a new cultured meat substitute for foie gras is presented.⁴ A main focus of the study is on how one can methodologically elicit willingness-to-pay for a product that is not yet commercially available, but the authors also find in a pre-post comparison that students’ pro-meat attitudes diminish after they learn about the foie gras alternative. In contrast, we experimentally manipulate both whether participants learn about cultured meat (as opposed to a plant-based or conventional farm meat product) and the appeal of the new cultured meat product. We also collect data from a representative U.S. sample and the product we consider is one that consumers may not feel to be as ethically problematic as foie gras: a meatball. Our experimental sample is quite large, enabling us to consider what one might *a priori* expect to be smaller effects in this broader population. In support of our main results, we also present findings from a similar experiment run on a very different sample: Amazon’s Mechanical Turk (MTurk). The latter results are somewhat smaller in magnitude but broadly consistent.

Our experiment also relates to the literature on motivated beliefs and cognitive dis-

⁴They also run a hypothetical version of the experiment on a nationally-representative sample of 1,200 people in France.

sonance (see, e.g., Bénabou and Tirole 2016 for a survey). Our finding on the effect of a positive view of cultured meat on moral beliefs is in line with the idea that consumption of a product can affect consumers’ beliefs about it (*e.g.*, Bénabou and Tirole 2016, Rabin 1994). In contrast, our finding on the effect of exposure to cultured meat relative to a conventional farm meat product suggests that there may be a backlash effect when participants do not like a new product. Building on a literature from psychology (Brehm and Brehm 2013) on a type of backlash effect, Boissonnet and Ghersengorin (2022) characterize a choice pattern in which consumers “protest” a perceived reduction in their choice set which would be consistent with our results, though other explanations are possible. Our setting also enables us to consider whether a plant-based substitute, which consumers may see as less threatening to their existing options, has the same effect. It does not, suggesting that changes in beliefs may depend on attributes of the new technology.

Similarly, this work relates to a broad literature on how people revise their positions when exposed to new information. For example, Schwardmann et al. (2022) show evidence of self-persuasion in international debating competitions, where individuals persuade themselves of a position’s superiority because they are required to devise arguments for it. Although we do not explicitly ask respondents to devise arguments to support their meat consumption, this self-persuasion dynamic may be triggered when respondents are exposed to a new product alternative. In another related vein, Taber and Lodge (2006) and Kashner and Stalinski (2024) show that individuals treat information differently depending on whether it accords with their pre-existing view or political party’s line. Our results are generally consistent with this pattern: because the large majority of respondents consume conventional farm meat, merely informing respondents about the existence of cultured meat may trigger scrutiny of the new alternative in order to defend their pre-existing beliefs. Prior literature also finds that motivated beliefs can lead people to avoid information (Serra-Garcia and Szech 2021, Huck, Szech and Wenner 2015). We find evidence of information avoidance in our setting, as well.

Finally, this study contributes to the broader literature on how technological change

and economic growth affect morality and social norms. Some work has suggested that technologies in widespread use shape social attitudes, as in the case of agricultural practices favoring exclusive male employment or greater time patience (Galor and Ozak 2016, Alesina et al. 2013). With regard to animal welfare in particular, several recent technologies have rendered previous practices toward animals obsolete and, in some quarters, repugnant: improvements in synthetic materials may have changed social attitudes towards fur coats, and fossil fuels eliminated the need for the use of whale oil (Shapiro 2018). Our study shows that the effect of a new technology on moral attitudes critically depends on key details surrounding how it is received.

2. Cultured Meat: A New Technology on the Horizon

In the final week of 2020, the Singaporean restaurant 1880 served just over 40 customers chicken meat produced without slaughtering a chicken for \$23 per person, a first for a business open to the public (RTN 2021, Business Wire 2020). This product followed on the heels of a competitor cultured meat product that was served for free in a promotion in Israel in October 2020 (Waltz 2021). Cultured meat regularly attracts great media attention and public excitement (*e.g.*, Bray 2021, Browne 2021). Despite these recent developments, however, cultured meat is off the market for nearly all consumers.

Cultured meat products come from actual animal tissue cells, an innovation over the plant-based sources that constitute other meat alternatives (*e.g.*, tofu, tempeh). While the production process continues to be refined, it involves taking a small number of cells from a living animal, producing a cell line, and cultivating the cells into a tissue using a bioreactor-fed protein, carbohydrates, and other nutrients (Waltz 2021). Cultured meat products' typical marketing sells them as a close substitute to conventional meat products in terms of nutrition and taste. For this reason, the availability of cultured meat products could reduce the perceived cost, including non-monetary costs, of avoiding conventional

animal products.

Cultured meat could lower farming’s environmental footprint. The Food and Agriculture Organization estimates that animal agriculture causes 14% of greenhouse gas emissions (Gerber et al 2013). Shifting to cultured meat products could drastically reduce such emissions. Current estimates suggest cultured meat requires 99% fewer greenhouse gas emissions, 96% less water and 99% less land than conventionally produced meat (Tuomisto and Teixeira de Mattos 2011, Tuomisto and Roy 2012, Mattick et al. 2015). Further, in the wake of the COVID-19 pandemic, there has been more public discussion of the potential impacts of animal agriculture on human health (Bernstein and Dutkiewicz 2021, Garcés 2020). Animal agriculture can not only lead to new infectious diseases, but the antibiotics fed to animals can lead to antibiotic resistance (Shea 2003, Smith 2002).

Cultured meat also has the potential to reduce intensive agriculture, which generates substantial public concern around animal welfare. Global food and agriculture producers slaughtered around 72 billion land animals in 2018 and 150 million tons of aquatic animals in 2013 (Our World in Data 2021a, Our World in Data 2021b). In the U.S., around 98.7% of animals on farms are on what the U.S. Environmental Protection Agency defines as “Confined Animal Feeding Operations” (CAFOs) (Anthis 2019). Animals on CAFOs frequently suffer health problems related to the intensive production, such as bone weakness and skin lesions for birds; bruising, disease, and lameness for cattle; and stress, food deprivation and poor water quality for fish (Ashley 2006, Bessei 2006, Grandin 2018). Surveyed members of the public express substantial concern for animal welfare, with consumers willing to pay 10-30% more for higher-welfare animal products (Grethe 2017, Johansson-Stenman 2018). Since cultured meat production does not require farm animals,⁵ animal suffering on farms would decline substantially.

As important as the direct substitution effects of cultured meat products might be, the indirect effects may be substantial. If new products raise moral and environmental concerns

⁵Currently, some starter serums used in production are derived from animals, however, the same serum can be re-used without additional animals being harmed. This serum thus can be thought of as similar to the inputs to many vaccines which were originally developed using animal cells. For example, the flu vaccine Flucelvax uses a cell culture derived from a cocker spaniel in 1958 (Ledford, 2012).

by alleviating cognitive dissonance, this could lead to a cycle of further welfare reforms and public support for cultured meat.

3. Experimental Design

Our main study sample consists of 6,000 U.S. residents sampled to be representative of the total U.S. population through the polling firm YouGov. A second set of experiments, conducted on Amazon’s Mechanical Turk, support the main results. The designs of the studies are broadly similar, so we describe the design of the YouTube study in this section and the MTurk studies in Online Appendix C.⁶ The study is pre-registered with the AEA RCT registry (AEARCTR-0001841), with pre-analysis plans posted to the Open Science Framework (OSF).⁷

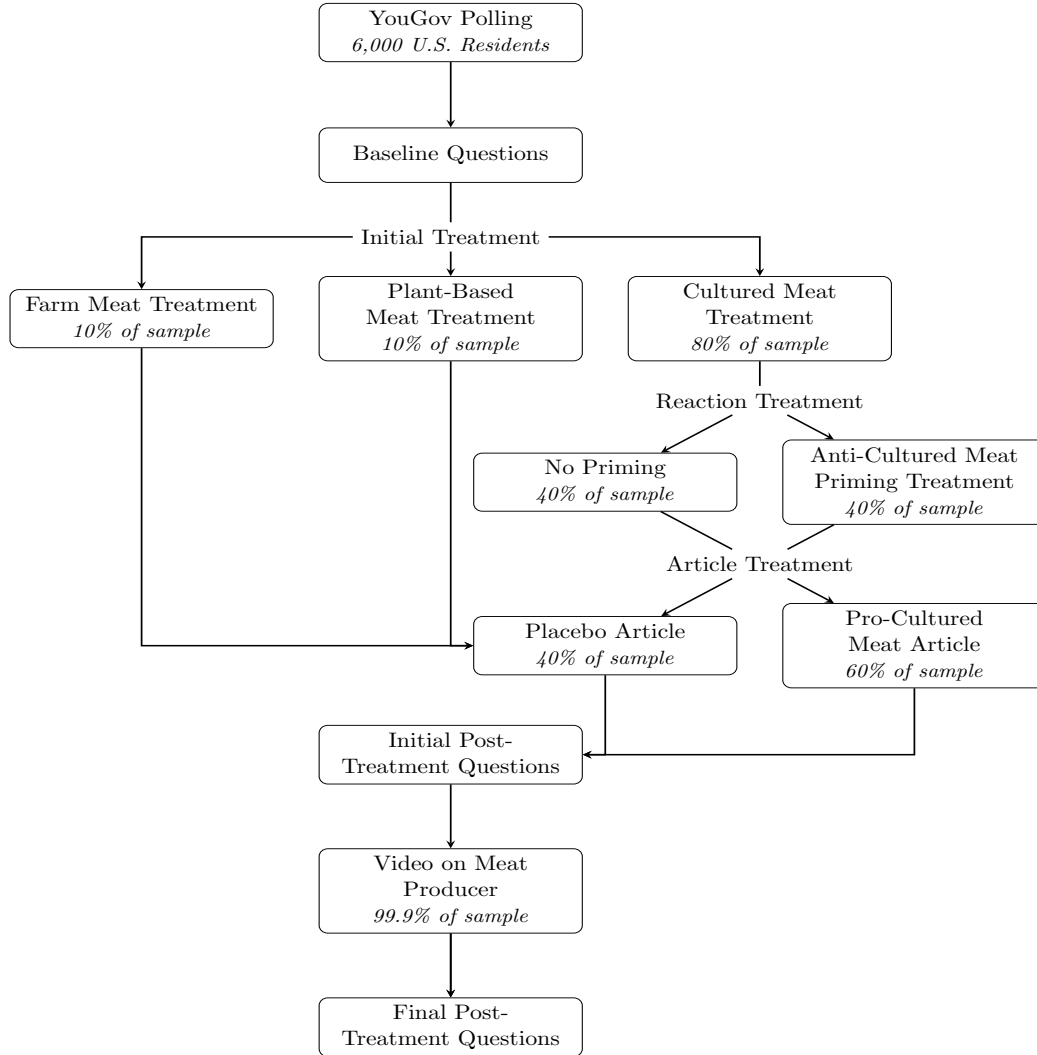
In the initial questionnaire, respondents answer a few sets of questions. First, respondents answer baseline questions before treatment assignment. Second, participants learn alternatively about a new cultured meat, plant-based meat, or farm meat product, and follow-up questions elicit immediate reactions to the new product. Those randomly assigned to view information about a cultured meat product are further randomized into receiving or not receiving further information treatments to manipulate how positively they feel about it, while those in the plant-based meat or farm meat arms receive a placebo article. Participants then view a video clip documenting animal abuse at a major supplier to Walmart, so that we can measure participants’ beliefs about animal welfare given actual exposure to concerns around conventional farm meat production. Finally, participants answer concluding questions including incentivized product and donation choices.

Figure 1 illustrates this design. We describe each set of questions in more detail below.

⁶The surveys have some small differences but include the same basic interventions and attitude measures. Both surveys include baseline questions about consumption and moral attitudes, exposure to a cultured meat alternative, a video investigation of a major pig farm, endline questions about consumption and moral attitudes, and questions regarding willingness to donate to an animal advocacy organization and relative valuation of farm and plant-based meat. Screenshots of the questionnaires appear in the Online Appendix.

⁷In particular, the pre-analysis plan for the main YouGov experiment is available at <https://osf.io/58u49>. It references a preprint based on the second MTurk experiment: https://doi.org/10.31219/osf.io/4dczh_v1, with the addition of eliciting incentivized WTP measures for plant-based and farm meat products. Pre-analysis plans for the earlier MTurk experiments are also available on OSF as part of the same project.

Figure 1: YouGov Experimental Design



Notes: The pro-cultured meat article is one of three: an article arguing that naturalness is meaningless and not necessarily good (“debunk unnaturalness”), an article arguing that many good and popular things are unnatural (“embrace unnaturalness”), or an article claiming that many people are interested in cultured meat. The treatments are described in a companion paper (Macdonald and Vivalt 2017) but none has any substantive effects (Appendix Table A16). The placebo article is an article on the benefits of reading. Within the cultured meat arm, the priming treatment (priming vs. no priming) and article treatment (placebo vs. pro-cultured meat) are fully cross-randomized.

3.1. Baseline Questionnaire

Prior to treatment assignment, respondents enter basic information about themselves and complete a food frequency questionnaire that asks them to indicate how often they eat each of a list of food categories (*e.g.*, dairy, chicken, fruit). They then indicate on Likert scales their level of agreement with the claim that eating vegetarian food is morally preferable to eating meat via factory farming and their levels of belief and concern that conventional meat production hurts animals and hurts the environment.

3.2. Treatments

Following the baseline questions, all respondents read two paragraphs describing the increase in meat consumption in industrialized countries and concerns around its toll on animals, environmental resources, and public health. We then randomize respondents into reading an article about a farm meat product (10% of sample), a plant-based meat product (10%), or cultured meat product (80%).⁸ The product in question in all treatment arms is a meatball.

All respondents then answer an initial set of post-treatment questions regarding their interest in the meatball product they viewed (farm meat vs. plant-based meat vs. cultured meat) and whether they would eat it.

For respondents assigned to read about the plant-based or cultured meatball, the main purpose of these paragraphs is to inform respondents of an alternative to conventional meat, rather than to manipulate the appeal of these alternatives.

Among the 80% of respondents assigned to learn about cultured meat, we then randomize respondents into two further treatments that seek to manipulate the appeal of cultured meat. First, we assign half of respondents to read reactions to cultured meat from participants in an earlier study, which describe cultured meat as “disgusting” and “unnatural.”

⁸To improve pre-treatment balance on unobservables across treatment groups, we stratify the sample into six blocks prior to all treatment assignments based on the total servings of meat the participant eats per week (as reported in the baseline food frequency questionnaire).

We refer to this as the anti-cultured meat priming treatment or “reaction” treatment.⁹ Second, we assign half of respondents to read one of four articles: either a placebo article or one of three persuasive article variants arguing in favor of the benefits of cultured meat technology, which we call the “article” treatment.¹⁰ Participants in the farm meat and plant-based meat arms all read the placebo article as well.

Next, respondents learn that they will be shown a short, 90-second clip of an undercover investigation from a major meat producer. This video contains graphic footage of the narrow confinement, mutilation, and early mortality of animals at a large pig farm that supplies Walmart. If the perceived availability of an alternative to farm meat makes participants more receptive to information about potential harms of conventional animal agriculture, we might expect that those who received the cultured meat treatment to react more to the video, especially those who do not receive the anti-cultured meat priming.

Before the video appears, we inform participants that they may be selected to receive a \$20 bonus and ask them how much of the bonus they would be willing to give up to *avoid* watching the video. We use a Becker-DeGroot-Marschak mechanism to incentivize the choice. For the 0.5% randomly selected for the bonus, we draw a random number from 1 to 20. If the respondent enters a value at least as high as that random number, they receive \$20 minus that value as a bonus via YouGov’s payment system; all other participants view the video. Ultimately, 99.9% of participants watch the video, so we do not limit attention to those who watched the video in our analyses.¹¹

⁹Should cultured meat become commercially available, one may expect conventional meat producers to encourage people to reject it through similar approaches.

¹⁰These three persuasive article variations are described in a separate paper (Macdonald and Vivaldi 2017). None has any significant impact (Appendix Table A16).

¹¹If one thinks that the pathway through which information about a new alternative could affect moral beliefs is through a channel of participants being more receptive to information about potential harms, the video could be thought of as the source of information about potential harms, without which the information about the new product alone may be unlikely to have an effect. As very few individuals did not watch the video, we cannot test whether those who did not watch the video changed their views.

3.3. Concluding Questionnaire

After watching the video, respondents again give their responses to the questions asked earlier about interest in the meat product, moral preference for vegetarian food, and harm to animals and the environment. They also answer Likert scale questions regarding how likely it is that they reduce their meat consumption in the next few weeks and how easy it would be.

Finally, we ask participants two types of incentivized questions.

First, we elicit their willingness-to-pay (WTP) for farm meat relative to plant-based meat. We present participants with three successive choices between a certain amount of vegetarian ground meat or conventional ground meat to obtain an interval in which their indifference point lies, and we carry out these choices by sending respondents gift cards for the selected items. We inform participants that we will randomly select a subset of respondents and send them the voucher they choose in one randomly drawn question out of the three. Participants in the cultured meat arm complete an additional set of questions regarding their preference for cultured meat versus farm meat. We advise participants that this latter set of questions is hypothetical and unincentivized.

Second, we ask participants how much of a potential \$20 bonus they would like to donate to a charity that helps prevent harm to farm animals. Again, participants are informed that we will carry out the choices of a randomly selected subset of participant, to make their choices incentive-compatible.¹²

3.4. Incentivization

We include several incentivized measures of respondents’ attitudes and product preferences. Respondents choose how much of a real \$20 bonus to donate to an animal-welfare charity; ten respondents are randomly selected and their decisions implemented. Partici-

¹²Though we do not discuss it in detail in this paper, we conduct a follow-up survey one month after the baseline experiment. We ask the same questions about values, beliefs, willingness to pay for meat and meat alternatives, and willingness to donate to an animal charity as in the survey described here. For details, see Online Appendix D. We repeat the same incentivization procedure as in the initial survey.

pants also make a series of real choices between certain amounts of farm meat and plant-based meat. Online Appendix B details the questionnaire, which involves a series of binary choices between farm meat and plant-based meat. The amounts are chosen such that they correspond to the actual dollar amounts of Omaha Steaks gift cards or vouchers for Beyond Meat products (converted to dollars based on the price at the time of the survey), with the dollar values ranging from \$6.60 to \$26.97. We then mail twenty respondents either gift cards for Omaha Steaks or vouchers for Beyond Meat products based on one randomly selected choice.

4. Results

In this section we detail the results of our pre-registered analyses, which focus on the effects of exposure to the cultured meat article on moral preferences and willingness to pay for farm meat vs. plant-based or cultured meat. We first provide context for our results by documenting consumers' attitudes toward cultured meat and the strong effects of anti-cultured meat social priming. Next, we show that positive views of cultured meat increase concern for animal welfare, but mere exposure to cultured meat has a negative effect relative to exposure to a farm meat or plant-based meat product. An incentivized measure of information avoidance further supports this finding of a backlash effect. We show that the effects of cultured meat information and anti-cultured meat priming generate an increased preference for farm meat. Finally, we leverage our social priming treatment to show that if respondents viewed cultured meat positively, the negative effect of exposure to cultured meat would give way to a positive effect.

4.1. Attitudes toward Cultured Meat and Social Priming

Our first result is that respondents view cultured meat in a largely negative light, and the anti-cultured meat social priming some respondents randomly receive furthers these negative views.

The majority of participants in the cultured meat treatment arm have little interest in

the product, but they also expect they would eat it. In the first row of Figure 2, we see that nearly 40% of respondents say they are “not interested at all” in cultured meat, and it seems more people are in either tail than for farm meat, already suggesting it might be polarizing. At the same time, a majority of respondents say they would eat the product, though a larger share of them would eat the farm meat or plant-based meat product.

We will use the negative social priming to instrument for how positively participants feel towards the new product by measuring the change in positivity towards the cultured meat product before and after the randomized social priming. The social priming is a good instrument for changes in positivity towards the new product if it meets both the relevance criterion and exclusion restriction. With respect to the relevance criterion, the second row of Figure 2 shows the broadly negative effect of the prime.¹³ Table 1 further displays the first-stage treatment effect of anti-cultured meat priming on the change in this positivity index. We again see a strong effect in the expected direction.

To meet the exclusion restriction, the instrument should not affect moral beliefs about animals except through changing how good an alternative participants consider cultured meat to be for farm meat. It is difficult to think of any reason why social priming against cultured meat could affect moral beliefs about animals except through changing opinions about how good an alternative it is: The social priming only refers to cultured meat and does not reference any other product, and the priming consists merely of real statements expressing dislike of the product from participants in a past experiment.

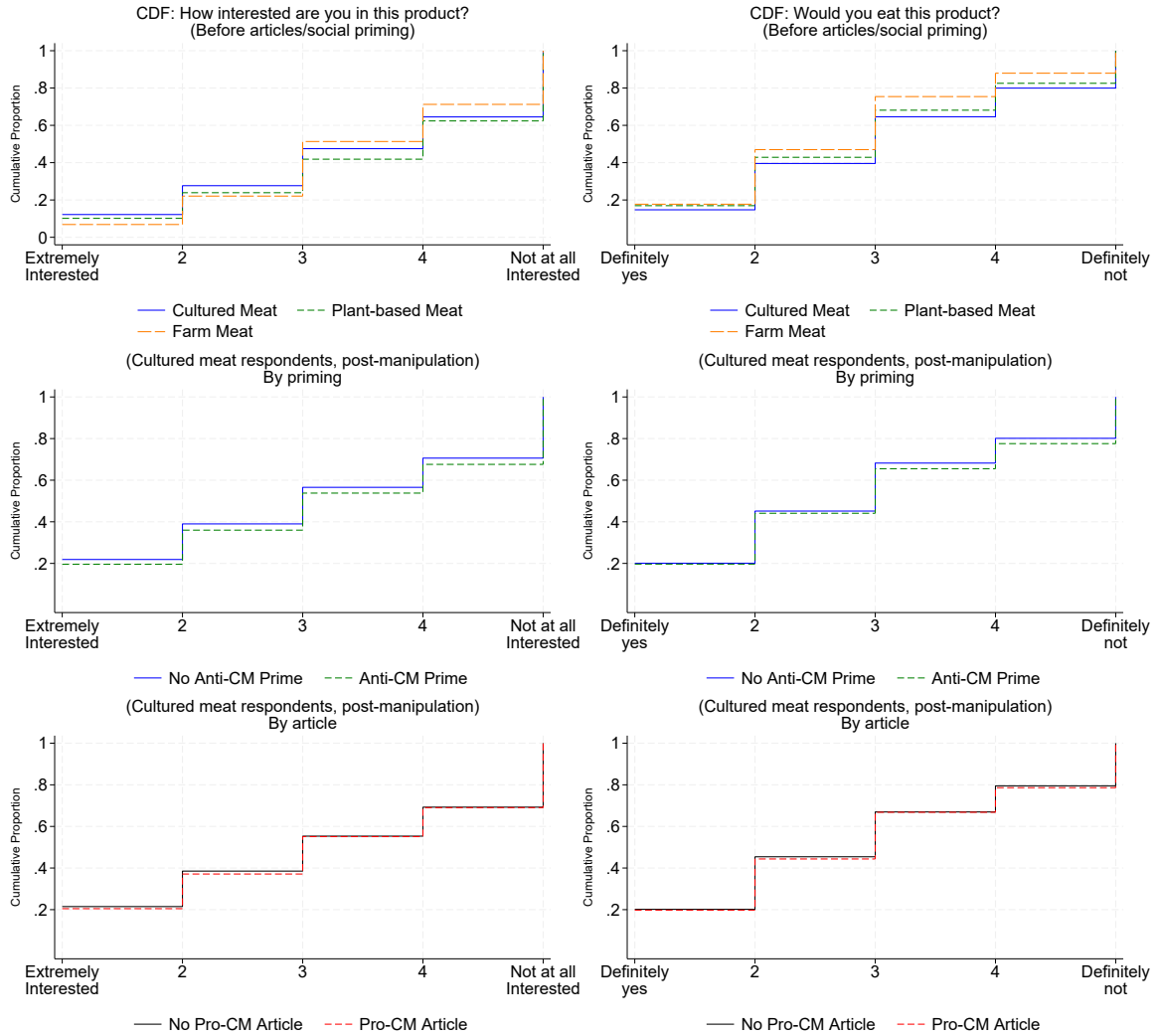
4.2. Moral Attitudes and Consumption

We now present our main results. We first show that exposure to information about cultured meat leaves respondents significantly less morally concerned by information about animal abuse in conventional farming than exposure to either information about farm meat or plant-based meat.¹⁴ Table 2 shows the raw means of key outcomes by treatment arm. There are no significant differences in beliefs between the treatment arms pre-treatment,

¹³The third row shows the striking lack of an effect of the pro-cultured meat articles.

¹⁴The pro-cultured meat articles, by contrast, have no effect (Appendix Table A16).

Figure 2: Respondents Are Not Interested in Cultured Meat but Expect They Would Eat It



Notes: The left subfigures plot the share of respondents exposed to cultured meat that express interest in a cultured meat product before additional treatments on a five-point Likert scale from “Not interested at all” to “Extremely interested.” The right subfigures plot the share of respondents who say they would eat a cultured meat product on a five-point Likert scale from “Definitely not” to “Definitely yes.”

Table 1: First Stage: Regression of Positivity and Easiness Indices on No Negative Priming Treatment

	(1) Change in Positivity toward Cultured Meat	(2) Easiness index
No Anti-Cultured Meat Priming	0.227*** (0.04)	0.219** (0.10)
Positivity toward Cultured Meat (Pre-Video)	-0.095*** (0.01)	0.401*** (0.02)
Observations	4812	4812

Notes: This table shows the effect of not receiving anti-cultured meat social priming on how positively respondents view plant-based eating and how easy they believe it would be to reduce their meat consumption. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Easiness is measured by the sum of two seven-point Likert scale questions assessing how easy respondents think it would be to eliminate farm meat from their diet and to reduce it (after flipping the scale so that higher is better).

showing that the sample is balanced. While moral concerns for animals appear higher in all groups post-treatment, the change is smallest for those shown information about cultured meat, particularly those who received a negative social prime about it.¹⁵

Table 3 displays regression results for the effects of the treatments on the view that eating vegetarian is morally preferable (Columns 1-3) and the amount of money out of a (real-stakes) \$20 bonus they would donate to a farmed animal protection organization (Columns 4-6). We present results from OLS regressions first, followed by results leveraging the exogenous instrument of anti-cultured meat priming. For those measures where we have baseline values, we always report the change in baseline values unless otherwise noted, so Columns 1-3 represent effects on changes in participant beliefs in whether eating vegetarian is morally preferable on the 7-point Likert scale.¹⁶ Receiving information about cultured meat lowers respondents' agreement that eating vegetarian is morally preferable by around a quarter of a point, relative to receiving information about farm meat (Column 1). While the point estimate for plant-based meat is also negative, it is smaller and not significant. We do not see significant effects of these treatments on the incentivized donation question (Column 4), though point estimates remain negative.

¹⁵As described earlier, these post-treatment questions were asked after participants watched a video about animal abuse at a farm supplying Walmart, so we would expect concerns to go up.

¹⁶I.e., the outcome could range from -6 to 6, as someone could change from a 1 to a 7 or vice versa.

Table 2: Raw Means of Outcome Variables, by Treatment Arm

	Farm meat	Plant-based meat	p-value (vs. farm meat)	Cultured meat, no prime	p-value (vs. farm meat)	Cultured meat, neg. prime	p-value (vs. farm meat)
Pre-treatment							
Morally preferable (1-7 scale) ^R	4.164	4.139	0.806	4.116	0.555	4.142	0.789
Belief about harm (1-5 scale) ^R	2.848	2.858	0.892	2.848	1.000	2.840	0.895
Harm is important (1-5 scale) ^R	3.221	3.266	0.537	3.265	0.443	3.287	0.258
Post-treatment							
Morally preferable (1-7 scale) ^R	3.393	3.456	0.562	3.577	0.035**	3.639	0.004***
Belief about harm (1-5 scale) ^R	2.097	2.142	0.518	2.153	0.323	2.211	0.047**
Harm is important (1-5 scale) ^R	2.418	2.512	0.225	2.524	0.082*	2.554	0.025**
Expects to reduce (0-10 scale)	5.547	5.320	0.261	5.277	0.089*	5.065	0.002***
How much reduce (0-10 scale)	5.119	4.988	0.507	4.912	0.184	4.713	0.009***
Donation (dollars)	9.983	9.719	0.587	9.555	0.264	9.539	0.247
Log WTP (for farm vs. plant-based meat) ^R	0.001	0.058	0.436	0.068	0.244	0.180	0.002***
Log WTP (for farm vs. cultured meat) ^R	N/A	N/A	N/A	-0.175	N/A	-0.006	N/A

Notes: This table presents mean values, pre- and post-treatment, for the main outcome variables. The pre-treatment values are prior to assignment to the farm meat, plant-based meat, or cultured meat treatment arms, while the post-treatment values are from questions asked following the video. The maximum and minimum values of the Likert scales are as follows: for whether eating vegetarian is “Morally preferable”, 1 = strongly agree, 7 = strongly disagree; for whether animals are harmed in conventional meat production (“Belief about harm”), 1 = a great deal, 5 = not at all; for whether this harm is important (“Harm is important”), 1 = a great deal, 5 = not at all; for whether the participant “expects to reduce” their meat consumption, “0 = definitely not, 10 = definitely yes”; for how much they expect to reduce it (“How much reduce”), 0 = not at all, 10 = a lot. The full Likert scales are available in the appendix. Raw values are presented, but those outcome variables for which a high value represents low concern for animals or stronger preferences towards farm meat have a superscript ^R next to them indicating this. Whether participants expected to reduce their consumption of farm meat and by how much, as well as their donation and WTP for farm meat vs. plant-based meat, was only elicited post-treatment. WTP for farm meat vs. cultured meat was only elicited from those who were in the cultured meat treatment arm, and hence comparisons cannot be made with those in the farm meat arm.

Table 3: Regression of Change in Beliefs and Donations on Treatments

	(1) Morally preferable OLS	(2) Morally preferable OLS	(3) Morally preferable 2SLS	(4) Donation (in dollars) OLS	(5) Donation (in dollars) OLS	(6) Donation (in dollars) 2SLS
Cultured Meat	-0.237*** (0.07)			-0.552 (0.43)		
Anti-Cultured Meat	-0.043 (0.04)			-0.014 (0.24)		
Priming	0.022 (0.04)			0.163 (0.28)		
Pro-Cultured Meat	-0.082 (0.08)			-0.268 (0.48)		
Change in Positivity toward Cultured Meat		0.247*** (0.01)	0.234 (0.15)		0.624*** (0.08)	0.227 (1.04)
Positivity toward Cultured Meat (Pre-Video)		0.169*** (0.01)	0.168*** (0.02)		0.705*** (0.05)	0.667*** (0.11)
Observations	6000	4812	4812	6000	4812	4812

Notes: Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents view plant-based eating as morally preferable and willingness to donate to an animal charity out of a \$20 bonus, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption, and Columns (1)-(3) control for the baseline value of the outcome variable. As described in the main text, where possible we consider changes in outcome variables pre-treatment and post-video; in this table, this is possible for Columns (1)-(3) but not (4)-(6).

A key driver of respondents' reaction is how positively they view cultured meat. Columns (2) and (5) of Table 3 show the correlation between the relevant outcome and how positively respondents view the product, while Columns (3) and (6) instrument for how positively respondents' view the product with whether they are exposed to the anti-cultured meat social priming treatment. We see that more positive views toward cultured meat are associated with increased moral preferences for avoiding farm meat and increased donations. The instrumental variables estimate is similar in magnitude for the moral preference Likert scale question but not significant.

Two more pieces of evidence show that greater receptivity to cultured meat increases concern with farm meat consumption. First, Table 4 considers beliefs participants have about harm to animals and the environment caused by farm meat production. In the first panel, respondents are asked how much they think animals are harmed and how important

they think this harm is. Respondents report significantly greater harms to animals when experimentally induced to feel more positively towards cultured meat. However, this does not translate to increased weights on the importance of this harm, and there are few effects on environmental attitudes. Second, Table 5 shows that the willingness of respondents to pay to avoid watching the video about factory farming is significantly lower after the anti-cultured meat priming. If participants experience a backlash effect, they may not be concerned that the information in the video would cause them to change their consumption, and the negative priming could make them feel more secure in this belief.

Turning to our consumption measures, we first present an unincentivized measure of how consumption responds to the informational treatments (Table 6). We see that those exposed to the anti-cultured meat priming are less likely to expect to reduce their consumption of farm meat (Column 1) and expect to reduce it by less (Column 4). There is again a negative effect of exposure to cultured meat alone compared to exposure to a farm meat product, although the effect is not significant. As in other cases, positivity toward cultured meat strongly predicts attitudes (Columns 2 and 5), and the results instrumenting for positivity support that this relationship is causal (Columns 3 and 6).

Incentivized, real-stakes consumption measures confirm that positive attitudes toward cultured meat have the effect of increasing moral concerns with conventional animal farming, though information about cultured meat itself does not. Table 7 displays the effects of the treatments on relative willingness to pay for farm meat compared to plant-based meat. Exposure to cultured meat information alone increases the relative value of farm meat compared to plant-based meat by around 10%, significant at the 10% level (Column 1), relative to exposure to farm meat. Exposure to the anti-cultured meat priming has a roughly equal effect significant at the 1% level for both plant-based meat (Column 1) and (hypothetical) cultured meat (Column 4). Again, positivity is strongly associated with willingness to pay (Columns 2 and 5), and these effects are causal (Columns 3 and 6).¹⁷

¹⁷Recall that only those provided information about cultured meat can answer questions about their WTP for it, hence the sample in Columns 2-6 is restricted to those exposed to cultured meat.

Table 4: Regression of Belief Change and Donations on Treatments

Animal-related outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Cultured Meat	-0.061 (0.05)			-0.081 (0.05)		
Anti-Cultured Meat	-0.062** (0.03)			-0.015 (0.03)		
Priming						
Pro-Cultured Meat	0.008 (0.03)			0.007 (0.03)		
Article						
Plant-based Meat	-0.040 (0.06)			-0.063 (0.06)		
Change in Positivity toward Cultured Meat		0.174*** (0.01)	0.295** (0.12)		0.190*** (0.01)	0.092 (0.12)
Positivity toward Cultured Meat (Pre-Video)		0.112*** (0.01)	0.126*** (0.01)		0.097*** (0.01)	0.086*** (0.01)
Observations	6000	4812	4812	6000	4812	4812
Environmental outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Cultured Meat	-0.019 (0.05)			-0.070 (0.05)		
Anti-Cultured Meat	-0.024 (0.03)			-0.017 (0.03)		
Priming						
Pro-Cultured Meat	-0.021 (0.03)			-0.005 (0.03)		
Article						
Plant-based Meat	-0.016 (0.06)			-0.059 (0.06)		
Change in Positivity toward Cultured Meat		0.159*** (0.01)	0.128 (0.11)		0.165*** (0.01)	0.098 (0.12)
Positivity toward Cultured Meat (Pre-Video)		0.104*** (0.01)	0.100*** (0.01)		0.095*** (0.01)	0.088*** (0.01)
Observations	5999	4812	4812	6000	4812	4812

Notes: Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents view animal farming as harming animals and the importance of that harm in the top panel, and analogous questions for the environment in the bottom panel, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption and the baseline value of the outcome variable.

Table 5: Regression of WTP to Avoid Information

	(1) WTP	(2) WTP	(3) WTP
Cultured Meat	-0.485 (0.36)		-0.160 (0.44)
Anti-Cultured Meat		-0.514** (0.24)	-0.514** (0.24)
Priming		-0.088 (0.28)	-0.088 (0.28)
Pro-Cultured Meat			
Article			
Plant-based Meat	-0.448 (0.49)		-0.448 (0.49)
Constant	9.075*** (0.44)	8.884*** (0.41)	9.075*** (0.44)
Observations	6000	4812	6000

Notes: Column (1) shows the effect of information about cultured meat and information about plant-based meat on respondents willingness to pay to avoid watching a video about factory farming. Column (2) shows the effect of anti-cultured meat social priming and an pro-cultured meat article on the same outcome. Column (3) shows all three effects.

Table 6: Regression of Change in Beliefs on Treatments

	(1) Expects to reduce consumption OLS	(2) Expects to reduce consumption OLS	(3) Expects to reduce consumption 2SLS	(4) How much expects to reduce OLS	(5) How much expects to reduce OLS	(6) How much expects to reduce 2SLS
Cultured Meat	-0.191 (0.18)			-0.151 (0.18)		
Anti-Cultured Meat	-0.212** (0.10)			-0.199** (0.10)		
Priming						
Pro-Cultured Meat	-0.106 (0.11)			-0.075 (0.11)		
Article						
Plant-based Meat	-0.227 (0.20)			-0.132 (0.20)		
Change in Positivity toward Cultured Meat		0.447*** (0.03)	1.058** (0.42)		0.398*** (0.03)	0.986** (0.41)
Positivity toward Cultured Meat (Pre-Video)		0.541*** (0.02)	0.599*** (0.04)		0.494*** (0.02)	0.551*** (0.04)
Observations	6000	4812	4812	6000	4812	4812

Notes: Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents expect to reduce their meat consumption and how much they expect to reduce it. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption.

Table 7: Regression of Relative WTP for Farm vs. Alternative Meat on Treatments

	(1) Log WTP (vs. plant-based) OLS	(2) Log WTP (vs. plant-based) OLS	(3) Log WTP (vs. plant-based) 2SLS	(4) Log WTP (vs. cultured) OLS	(5) Log WTP (vs. cultured) OLS	(6) Log WTP (vs. cultured) 2SLS
Cultured Meat	0.114* (0.07)					
Anti-Cultured Meat	0.111*** (0.04)			0.169*** (0.04)		
Priming						
Pro-Cultured Meat	-0.062 (0.04)			-0.032 (0.04)		
Article						
Plant-based Meat	0.058 (0.07)					
Change in Positivity toward Cultured Meat		-0.151*** (0.01)	-0.539*** (0.16)		-0.245*** (0.01)	-0.810*** (0.17)
Positivity toward Cultured Meat (Pre-Video)		-0.200*** (0.01)	-0.237*** (0.02)		-0.289*** (0.00)	-0.343*** (0.02)
Observations	5999	4811	4811	4812	4812	4812

Notes: Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on the logarithm of respondents' willingness to pay per unit of plant-based meat and cultured meat, respectively, in units of farm meat. Column (4) does not include assignment to information about cultured meat as an independent variable because willingness to pay for farm vs. cultured meat was only measured for respondents exposed to information about cultured meat. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption.

As a final demonstration of the importance of how respondents view cultured meat, we consider what the effects of information about cultured meat would be if participants viewed it positively. We simulate the reaction of a respondent to information about cultured meat by estimating a two-stage least squares regression where we instrument for how positively respondents are made to feel about the product using the social priming treatment. We then use the coefficients on cultured meat information and respondents' positivity to estimate the effect of cultured meat information for a respondent who is interested in and would eat cultured meat. Table 8 shows the results. Under these circumstances, there is a significant positive effect of providing information about cultured meat on moral concerns with conventional animal farming and reductions in both incentivized and unincentivized measures of farm meat consumption. These results illustrate that the largely negative effects of providing information about cultured meat on moral concern for animals are offset if respondents feel positively towards the new product.

4.3. MTurk Results

These findings replicate and extend earlier experiments run on MTurk in 2016-2017. The design of these experiments is described in Online Appendix C, but the first experiment, in 2016, finds that exposure to information about cultured meat tends to reduce moral concern for animals, with the effect driven by individuals who express negative feelings towards the product. These findings motivate a second experiment in 2017 in which we randomize receipt of a negative social priming arm, as in the main YouGov sample, to experimentally manipulate how positively participants feel towards the new product.

While this second experiment differs in several respects from the YouGov experiment that is the focus of this paper,¹⁸ its results still lend support to our main findings. In particular, Figure 3 plots the coefficients from the YouGov and MTurk samples' instrumental variables regressions, focusing on how the experimentally manipulated positivity towards cultured meat affects moral beliefs and expectations about future meat consumption. We

¹⁸For example, it uses a pure control, not informed about any new product, rather than a group informed about a farm meat product.

Table 8: Predicted Effects of Cultured Meat Information Given Positive Views

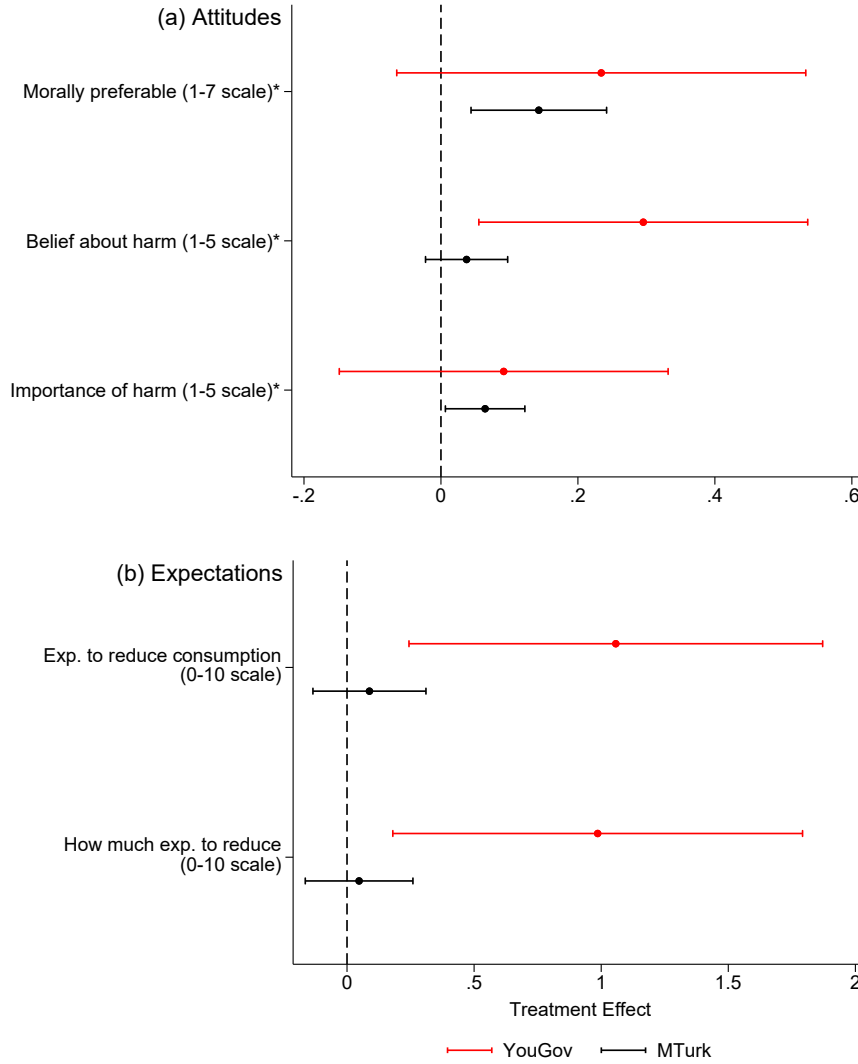
	Predicted Effect
Morally preferable	0.183 (0.47)
Donation	-0.262 (2.85)
Harms animals	0.683 (0.47)
Harm to animals is important	0.032 (0.37)
Harms environ.	0.261 (0.35)
Harm to environ. is important	0.073 (0.35)
Expects to reduce	2.119* (1.24)
How much reduce	2.026* (1.22)
Log WTP (for farm vs. plant-based meat)	-1.196** (0.56)
Observations	5999

Notes: Each row shows the estimated effect of information about cultured meat combined with a simulated social prime that makes a respondent positively disposed to cultured meat. We use a two-stage-least-squares procedure to estimate the effects of cultured meat information and, via the social priming instrument, a positive view of cultured meat on each outcome. We then add the effect of cultured meat information to the effect of being positively disposed, defined as averaging a four out of five (after flipping the scale so that higher is better) on the two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it. The outcome variables are constructed as in the earlier regression tables, such that they represent changes between pre-treatment and post-video values wherever possible.

observe that despite the samples being very different, in both samples feeling more positively toward cultured meat (as instrumented by the social priming treatment) results in participants placing more moral weight on animals.¹⁹ Full results are provided in Online Appendix Tables A1-A10.

¹⁹A few differences between the MTurk sample and the YouGov sample may be worth noting. First, as described earlier, the YouGov sample is a nationally representative sample; while the MTurk sample is restricted to participants in the US, those taking surveys on MTurk tend to be lower-income on average than in the YouGov sample. Second, despite attempts to ensure attentiveness through attention checks, one may expect that the MTurk sample, being paid less, may put less effort into answering questions accurately, biasing results towards 0. Third, given the lower cost of running experiments on MTurk, many animal welfare organizations have run experiments on this platform, while few had used YouGov at the time of our experiment. From free text responses, it is clear that some MTurk participants had previously participated in online surveys relating to animal agriculture and may have formed stronger views about animal welfare before the experiment; in particular, some participants expressed views consistent with having previously been polarized against having concerns for animal welfare. Overall, we expect these factors would tend to weaken the MTurk results relative to the YouGov results, though we still observe significant impacts in the same direction.

Figure 3: Comparison of Main MTurk and YouGov Results



Notes: This figure provides a comparison of the main results about moral concerns towards animals in the YouGov and MTurk samples, drawing from the 2SLS coefficients from Tables 3, 4 and 6 and Appendix Tables A7-A9, respectively. Recall that we use the change in attitudes post-treatment, compared to pre-treatment, as our preferred measures, for those for which we have both pre-treatment and post-treatment values (indicated here with an asterisk). While donations were also elicited in MTurk, these were elicited out of a much smaller potential bonus amount (10 cents as opposed to a chance of a \$20 bonus) and participants may have thought of these bonuses differently within these different platform ecosystems, so we do not think these results comparable.

5. Discussion

The results we observe - a negative impact of exposure to cultured meat on concerns for animal welfare but a positive impact driven by positive attitudes - call for an explanation. It is puzzling that a substitute that a participant is not interested in could affect their beliefs given that they can presumably choose not to eat it.

There is suggestive evidence that a backlash effect could be part of the explanation. More specifically, respondents may view the arrival of cultured meat as signifying the undesirable replacement of farm meat by cultured meat. This is consistent with the data: participants in the cultured meat treatment arm are less likely to say they would eat the product than those in the farm meat or plant-based meat treatment arms (Figure 2). Respondents may therefore feel incentivized to fight more for farm meat, or they may simply react against a perceived attack. That the positivity of participants' view of the alternative mediates the effect is also consistent with this explanation.²⁰

It is notable that merely being informed of a plant-based alternative does not provoke this negative reaction. This suggests that the effect is tied to cultured meat as a new technology. Participants are already familiar with plant-based meat alternatives and can observe that they have not replaced farm meat or substantially limited their options. They may have more reason to fear that cultured meat might replace farm meat.

Another possibility one might consider is whether those in the cultured meat arm update positively on farm meat in some way. For example, information about cultured meat might lead respondents to expect that in the future, the negative externalities of farm meat production on the environment or on animals will improve through technological progress. We think that this explanation is less likely because the questions about whether animals experience harm through conventional meat production and how much harm animals experience are explicitly about current meat production rather than future meat production.²¹

A more general possibility is that respondents view cultured meat as a type of meat and

²⁰For a theoretical investigation, see Online Appendix A.

²¹E.g., "How much do you think conventional meat production, i.e. factory farming, harms animals?"

develop more positive attitudes toward all meat in response to information about it given its lesser impact on animals and potentially the environment. For example, participants might consider the space of options as roughly consisting of two bins: farm meat and plant-based meat. In this case, cultured meat may fall in the farm meat bin. This explanation is somewhat difficult to reconcile with the secondary results. For example, this explanation would predict that, compared to information about farm meat, information about plant-based meat would have an effect that is directionally opposite to that of information about cultured meat. This explanation would also predict that anti-cultured meat priming would have the reverse effect to what we see in the data.

Another possibility is that those in the farm meat arm become more concerned about animals after watching the video precisely because it relates more directly to the product they just learned about - another farm meat product. For some people, it may be the first time they are encountering a video of an undercover investigation and it could affect them more if they had just been thinking of a farm meat product that they anticipated liking. This would not explain why those in the plant-based arm often change their views by more than those in the cultured meat arm, so this could not explain our full set of results. However, we cannot rule out this explanation contributing somewhat, e.g., if the sample is comprised of several different types of participants with different reactions.

Ultimately, we do not want to lean too much on any one particular explanation. “Reactance” or a backlash effect, driven by those who feel negatively towards cultured meat, seems to be the explanation that is most consistent with all our findings.²² But, regardless of the explanation, our results suggest that cultured meat is not a panacea for animal

²²The responses to free text entry questions also broadly support this hypothesis. For example, when those in the cultured meat arm are asked for their initial reactions, before the reaction or article treatments, some immediately defend farm meat, e.g.: “I reject this idea. Farming and ranching has been around forever. I will continue to eat as much meat as I can”; “I’m all for real animals being slaughtered for eating on mass scale. Clean meat nope I will not eat [this] so called meat... it’s not natural or real”; “Sounds too fake. We are above the animals and they need to be slaughtered.”

For others, thinking about cultured meat may make them reflect on how much they like farm meat, e.g.: “I prefer to eat grass fed meat and organic”; “I will continue to eat beef”; “I would not eat conventional farm raised nor lab created meats. I buy locally from farmers that are organic or mostly organic.”

Any of these reactions could nudge participants to feel relatively more positively towards farm meat and be less open to considering arguments against it.

welfare and environmental harms of animal agriculture.

Overall, we do not see much evidence that cultured meat leads to a cycle of greater concern about conventional farm meat production. It may paradoxically even lead some people to care relatively less about animal welfare. However, given that these effects are driven by how positively participants treat the new alternative, we cannot rule out that an alternative that participants felt much more positively about would lead the average participant to care more about animal welfare.

Finally, we follow up with the participants a month after the initial experiment and find surprisingly durable effects of the relatively light-touch treatments. 81% of the initial sample responds to the follow-up survey. The effects on ethical beliefs and expected consumption largely remain (Online Appendix Tables A11-A15).

6. Conclusion

Consumers face a myriad of ethical choices, often selecting the less ethical option and resolving the resulting dissonance through rationalization or willful ignorance. In this study, we examine whether providing consumers with higher quality or more convenient ethical goods reduces these incentives for rationalization, leading consumers to select ethical options more frequently and soften the attitudes and beliefs they hold to justify the less ethical option.

In a representative online panel, we find that exposure to cultured meat lessens moral concerns with animal agriculture compared to exposure to farm meat or plant-based meat. An experimental nudge to view the product more negatively (as conventional meat producers might be expected to produce) exacerbates this effect. We show that positive views toward cultured meat appear to mediate its effect on moral attitudes. Participants' willingness to pay for farm meat, plant-based meat, and cultured meat is consistent with these findings, and some of the effects persist a month later. While the effect of informing participants about cultured meat on moral attitudes is relatively negative, suggesting there can be a backlash effect, our evidence suggests that if respondents viewed cultured meat

positively, it would increase moral concerns about animals. Our results highlight that the ultimate impacts of a new technology critically depend on how it is received.

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Online Appendix: The Impact of New Products on Ethical Beliefs

Appendix

Zach Freitas-Groff Bobbie Macdonald Eva Vivalt

A. Theoretical Framework

This section clarifies how a simple model of motivated beliefs would predict people respond to information about a novel meat alternative. The model begins from Hestermann et al. (2020) but adds to it the existence of a meat alternative.

Consider a consumer who plays a two-period signaling game with their future self as in Hestermann et al. (2020). In the first period ($t=0$), the consumer receives a signal m about the harms of factory farming and transmits a message to their future self. In particular, they either receive the signal that factory farming causes a little (m_L) or a lot (m_H) of harm. They then send a message $\hat{m} = m_L$ or $\hat{m} = m_H$ to their future self. In the second period ($t=1$), they receive this message and choose how much farm meat to consume (c), incurring a moral cost $\omega\tilde{x}c$ to eating meat that depends in part on how harmful they believe eating meat is to animals. The parameter ω represents individual intensity of guilt, and $\tilde{x}$ represents how harmful they believe factory farming is. In the second stage the consumer sets c^* according to:

$$c^* = \arg \max_{c \in \mathbb{R}_+} U(c) - p_c c - \omega\tilde{x}c$$

where p_c is the price of the farm meat and U captures their basic preference for eating meat. The value of $\tilde{x}$ is determined from the observed message by Bayesian updating, given the equilibrium strategy of the period-0 self. The period-0 self's utility, in turn, is assumed to equal the period-1 self's utility, minus a fixed self-deception cost k if the period-0 self chooses to send message m_L when the observed signal was m_H .

These payoff specifications define a game between the two selves. Hestermann et al. (2020) assume that the period-0 self reports good news truthfully ($\hat{m} = m_L$ if $m = m_L$), so that self-deception, if it arises, consists of misreporting bad news (setting $\hat{m} = m_L$ when $m = m_H$). The game has a unique equilibrium of this form, potentially using mixed strategies. Hestermann et al. (2020) then examine the comparative statics of this equilibrium. Among other results, they show that if consumers vary in their utility function U , consumers who have a stronger taste for meat will engage in more self-deception in equilibrium. An intuition is that the benefit of self-deception is higher for consumers whose meat consumption is higher.

We can enhance the above model to study what happens when a new product gets introduced. Introducing a new meat alternative, a , with price p_a and no moral costs, in the second stage the consumer will instead solve:

$$\max_{c, a \in \mathbb{R}_+} U(c, a) - p_c c - p_a a - \omega \tilde{x} c$$

Individuals again have the incentive to engage in self-deception. Now, however, the incentive is altered by the presence of the alternative, which can affect their farm meat consumption c^* . This allows us to make predictions about what happens when a new product is introduced.

If we consider the introduction of a new product simply as an expansion of the choice set to include a substitute for meat, then a new alternative should make people more willing to harbor concerns with factory farming. To see this, treat the introduction of a product as the lowering of that product's price from a prohibitively high one to a much lower one. If c and a are substitutes, "introducing" a lowers optimal consumption c^* . This decreases the payoff for self-deception.

On the other hand, consumers might see the introduction of a new meat alternative as a change in the available alternative or, if it replaces meat, a contraction in their choice set. If consumers believe that cultured meat would displace plant-based meats or even non-

meat options, this might lead them to expect they will consume more meat in the future. Separately, if consumers expect cultured meat will replace farm meat, they might view its introduction as the loss of an option. In the model presented in this section, not eating meat from animals reduces the motive or self-deception. Boissonnet and Ghersengorin (2022) present an alternative deviation from rational choice in which consumers “protest” against contractions in their choice sets. This sort of backlash (or “reactance”) effect could perversely increase the drive toward self-deception and lessen concern with factory farming.

B. Questionnaire

Participants answered the initial questions:

In this part of the survey, we’ll ask you about your food consumption. We know that what people eat often changes over time. They may have goals to eat some things more or less overtime. But right now, we are just curious about what you have eaten over the past month.

The image below shows the approximate size of a single serving for different categories of food.

Thinking about your diet over the past 30 days, please select the responses that best describe how often you eat a serving of each of the following types of food.

Options: Never, Less than 1 time per week, 1-6 times per week, 1-3 times per day, 4 or more times per day

- Dairy (cheese, milk, yogurt, etc.)
- Chicken (fried chicken, in soup, grilled chicken, etc.)
- Turkey (turkey dinner, turkey sandwich, in soup, etc.)
- Fish and seafood (tuna, shrimp, crab, etc.)

- Pork (ham, pork chops, ribs, etc.)
- Beef (steak, meatballs, in tacos, etc.)
- Other meat (duck, lamb, venison, etc.)
- Eggs (omelet, in salad, in baked goods, etc.)
- Fruit (apples, bananas, oranges, etc.)
- Vegetables (carrots, mushrooms, potatoes, etc.)
- Nuts (almonds, cashews, walnuts, peanuts, etc.)
- Beans (soy, chickpeas, chili, etc.)
- Veggie meats (tofu, veggie dogs, veggie burgers, tempeh, etc.)
- Grains (breads, pasta, rice, etc.)
- Other added fats (oil, salad dressing, chocolate, etc.)
- Sugary foods (pop/soda, candy bars, etc.)

Some people say that eating vegetarian food is morally preferable to eating meat produced via factory farming. What do you think?

Options: Strongly agree, Agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Disagree, Strongly disagree

How much do you think conventional meat production, i.e. factory farming, harms animals?

How concerned are you about this harm?

How much do you think conventional meat products harm the environment?

How concerned are you about this harm?

Options: A great deal, A lot, A moderate amount, A little, Not at all

Please read the information on the next page. Once you are finished, we will ask you several questions about it.

Participants then randomly saw one of the following pages of paragraphs on farm meat, plant-based meat, or cultured meat:

Figure A1: Treatment Arm 1: Farm Meat

Conventional meat products continue to be developed by many companies. Startups and scientists continue to develop ways of raising animals and producing meat. Scientists have been experimenting with new ways to breed and slaughter animals.

Doing so increases the supply of meat products over time. Meat consumption and its associated impacts will likely continue to grow in the future. Companies produce an array of meat products to sell to consumers.



Meatball

Figure A2: Treatment Arm 2: Plant-Based Meat

Vegetarian meat substitutes—that is, plant-based proteins with the shape and texture of meat—offer significant benefits addressing each of these problems, and seemingly-identical substitutes are currently being developed by several startup companies. Scientists aim to recreate meat from plants.

Doing so avoids slaughter, mutilation, and extreme confinement of farmed animals. Vegetarian meat also minimizes pollution and resource usage. Finally, companies can shield vegetarian meat production from disease and contamination without the use of antibiotics.



Meatball substitute

Figure A3: Treatment Arm 3: Cultured Meat

"Clean" meat—that is, real meat grown in a lab from animal cells without the need to raise and slaughter an actual animal—offers significant benefits addressing each of these problems, and is currently being developed by several startup companies. Scientists aim to grow meat from animal cells.

Doing so avoids slaughter, mutilation, and extreme confinement of farmed animals. Clean meat also minimizes pollution and resource usage. Finally, companies can shield clean meat production from disease and contamination without the use of antibiotics.



Clean meatball

Participants continued on to answer the following questions:

In 1-2 sentences, please state your main conclusion from the article you just read.

Which of the following did the paragraphs you just read NOT discuss (select multiple)?

Options [randomize]: Rising demand for meat; Religious objections to agriculture; Human health and disease; Laws to promote food safety

How interested are you in this product?

Options: Extremely Interested, Very Interested, Moderately Interested, Somewhat Interested, Not interested at all

Would you eat this product?

Options: Definitely yes, Probably yes, Might or might not, Probably not, Definitely not

Half of participants in the cultured meat arm randomly saw the following prime (3x):

Figure A4: Treatment Arm 3x: Anti-Cultured Meat Priming

A lot of consumers are turned off by clean meat. According to a survey in journal PLOS ONE reported by Live Science, no more than one-third of people would eat clean meat regularly, and many people are too grossed out to try it.

"On average, people see clean meat as... less natural, tasty and appealing," according to Matti Wilks, who authored the study.

Past participants in the current study have described clean meat as too unnatural for our digestive systems.

Here is a selection of comments that other survey respondents have made about clean meat:

"Artificial meat sounds disgusting."

"This seems very unnatural. I don't feel comfortable about this."

"No one will like clean meat."

"Clean meat isn't real meat."

"Our guts are not meant to digest unnatural things."

Please describe your feelings about clean meat:

All participants in the farm and plant-based meat arms saw an article about reading (3a), but participants in the cultured meat arm randomly saw that or one of three other articles (3b, 3c, 3d):

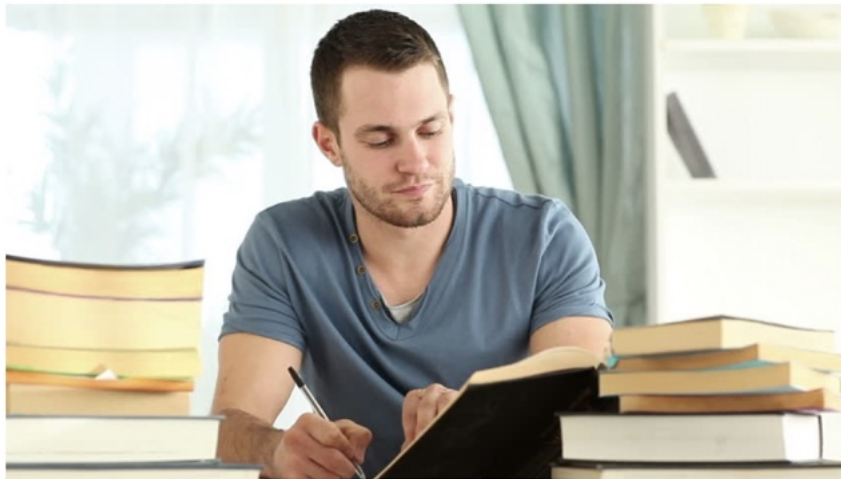
Please read the news article on the next page. Once you are finished, we will ask you several questions about it.

Treatment Arm 3a: Benefits of Reading

Daily reading can improve physical and mental health.

Books continue to be published by a range of companies and authors. Taking in this key part of our culture is not only fun but also yields many benefits. We've all been told to read from the time we're young, but research shows it may have more benefits than many of us thought.

[According to Mission.org](#), "It turns out that reading silently for only six minutes has been shown to reduce heart rate and ease muscle tension. In fact, that small amount of reading worked better for relaxation than other proven methods such as listening to music, going for a walk, or sipping a cup of tea.



Reading can decrease stress.



You can carry a book anywhere you go.

"Regularly reading just a bit more than that—only 30 minutes a day—has been shown to add two years to your life. That's right—developing a habit of reading for just a half hour per day can make you live years longer."

Treatment Arm 3b: Debunk Unnaturalness

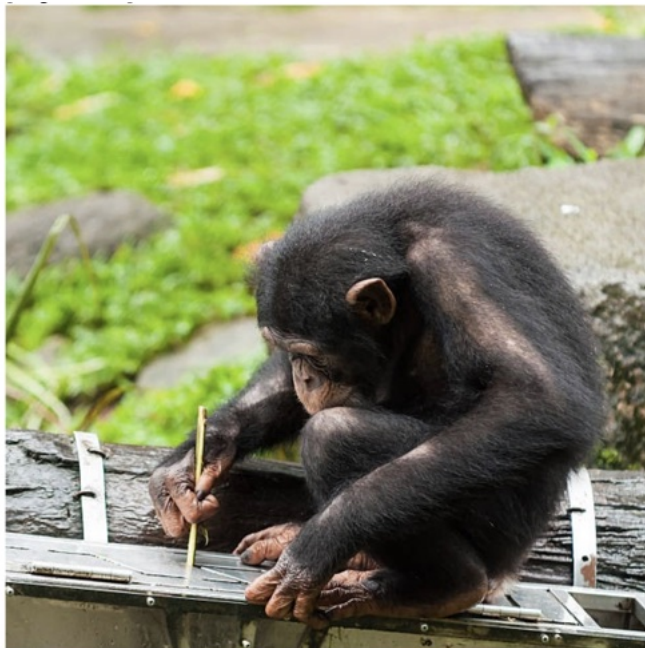
Natural doesn't mean good.

Are you turned off by a new product because it's not natural? It turns out that "natural" and "good" are two very different things.

Naturalness is a poor guide to most of what we care about, from health to safety to ethics. Each year, hurricanes have caused billions of dollars of damage in the U.S. and killed dozens or sometimes thousands of people. People around the world donate money to help those affected by hurricanes, and governments strive to stem the damage from hurricanes and other tropical storms. Have you ever paused to note that hurricanes are natural? Disease, bee stings, and tornadoes are all natural, but nobody would argue that any of these are good. Human societies go to great efforts to avoid many natural things.



Hurricanes are part of nature.



Chimpanzees use tools, so which tools are natural, and which are not?

Moreover, it's not even clear what "naturalness" means. Are tools natural since tool use played a role in human evolution? How sophisticated can a tool be before it stops being natural? Is having a shelter from the weather natural? Once we start thinking about what is natural and what is not, it is hard to draw a line.

Treatment Arm 3c: Embrace Unnaturalness

Many popular things are unnatural.

Think a new product is "unnatural"? Well, many of our surroundings and possessions are also unnatural, so it fits right in.

Iceboxes, one of the most mundane objects in today's world, were once seen as violating nature. Many were disturbed when people first began artificially freezing water using machines. According to the [Big Think](#), "Critics labelled [artificial ice] 'unnatural.' Slicing chunks of ice from lakes was the way to go, they said. Never mind the pollutants contained in the water. Machines, by contrast, freeze water that's been boiled and purified, making it a safer choice." Fast forward to today, and artificially-frozen ice is far more common in restaurants and households. Could you imagine if restaurants and home kitchens did not have ice boxes?



Most ice in households and restaurants comes from artificial machines.

Most ice in households and restaurants comes from artificial machines.



The unnatural process of brewing beer.

Many of our favorite foods and drinks are also unnatural. "As New Harvest, an organization that funds cultured meat research, states on its website, "bread, cheese, yogurt and wine are unnatural. All involve processing ingredients derived from natural sources,"" [Amy Fleming at the Guardian notes](#). The ranks of "unnatural" foods include yogurt, prized for its healthiness, and wine and cheese, synonymous with elegant meal.

So eat up! Clean meat joins the ranks of many essential and unnatural things.

Treatment Arm 3d: Descriptive Norm

Consumers look forward to buying clean meat.

[Engadget reports](#), "While there are still so many unknowns when it comes to cultured meat, there is evidence that people are generally on board. A [study published last year](#) in *PLOS One* found that 65 percent of 673 surveyed US individuals would probably or definitely try cultured meat while only 8.5 percent said they definitely would not. And adding to the evidence of public support, in 2016, Israel-based [SuperMeat](#) raised more than \$230,000 in an [Indiegogo crowdfunding campaign](#). In an email, SuperMeat co-founder [Shir Friedman](#) said, 'The crowdfunding was a way for [SuperMeat](#) to show potential investors, as well as the world, that there is massive public support for clean meat.'"

Moreover, demand for "clean" meat products keeps growing, despite higher price points compared with conventional meat, according to data from Nielsen Perishables Group. Clean meat offers an answer to a number of consumer concerns with the meat industry, including transparency, safety, and sustainability.



Surveys find strong consumer interest in clean meat.



Americans are increasingly concerned with food safety and transparency.

[A recent poll by Oklahoma State University](#) found 47% of Americans support a ban on slaughterhouses. Clean meat offers a way for consumers to have their ethics and eat their meat, too. Maybe that's why 72 percent of beef and pork eaters said they would eat clean meat.

In 1-2 sentences, please state your main conclusion from the article you just read.

Next, you will be presented with a short, 90 second video about factory farming. Some images may be disturbing.

Participants in this study will be entered in a lottery for the chance to win a bonus of up to \$20. How much of this bonus would you be willing to give up to avoid watching the video should you win? You may enter any number from \$0 to \$20. It is in your interest to honestly describe what you would pay to avoid watching the factory farming video.

Options: Integers 0-20

[DETAILS ONLY IF THEY CLICK:] As an example, if you would be willing to give up the entire bonus to skip the video, please enter 20. If you would be willing to give up \$1 of the bonus to skip the video, please enter 1.

Regardless of what value you enter, there is some chance you may be asked to watch the video and will be awarded the full bonus. However, this probability depends on what you enter, and if you enter a higher number you have a lower chance of being asked to watch the video.

More specifically, if we choose you to receive a bonus, we will pick a random number from 0 to 20 and charge you that number if it is less than the number you state in exchange for not showing the video. If the number is greater than the number you state, we will show you the video. Therefore it is in your interest to say the largest amount you would give up to avoid watching the video.

*Participants randomized to watch the video or not watch the video. Those assigned to the video see the following:*²

Thank you for your response. Please watch the following short, 90 second video. Following the video, you will be asked a few related questions. <https://youtu.be/9-Qq3CeMgg0>

Which of the following companies' practices does the video document (select multiple)?

Options [randomize]: McDonald's; Tyson; Walmart; Smithfield

Thank you for your response. In the next section, you will be asked a set of questions about your preferences and beliefs.

How interested are you in this product?

Options: Extremely Interested, Very Interested, Moderately Interested, Somewhat Interested, Not interested at all

Would you eat this product?

Options: Definitely yes, Probably yes, Might or might not, Probably not, Definitely not

Now please rate the following according to how difficult you think each goal would be for you.

Think about how much effort it would take, whether you feel like it is something you could possibly do, and whether it is something you could stick with over time.

[randomize order]

²Note that the video link has changed. The clip in question was from an undercover investigation by Mercy for Animals of a Tyson factory farm supplying Walmart (Mercy for Animals 2013). The specific video is called "Tyson Pork Group", dated February 1, 2013 on Mercy for Animal's website: <https://mercyforanimals.org/our-work/undercover-investigations/>. As of September 9, 2025, the video is available on Youtube at <https://www.youtube.com/watch?v=T48yOYjz5sk>. Approximately the first 90 seconds was shown (the clip started a second or two into the full video).

Completely eliminating conventional meat products from your diet in the next year

Reducing your consumption of conventional meat products by 25% in the next year

Options: Very easy, easy, somewhat easy, neutral, somewhat difficult, difficult, very difficult

Some people say that eating vegetarian food is morally preferable to eating meat produced via factory farming. What do you think?

Options: Strongly agree, Agree, Somewhat agree, Neither agree nor disagree, Somewhat disagree, Disagree, Strongly disagree

How much do you think conventional meat production, i.e. factory farming, harms animals?

How concerned are you about this harm?

How much do you think conventional meat products harm the environment?

How concerned are you about this harm?

Options: A great deal, A lot, A moderate amount, A little, Not at all

Do you expect to reduce your consumption of conventional meat products over the next few weeks?

Options: 0 to 10, 0 = Definitely not, 10 = Definitely yes

How much do you expect to reduce your consumption of conventional meat products over the next few weeks?

Options: 0 to 10, 0 = Not at all, 10 = A lot

Finally, respondents completed the following series of questions, with each indent indicating questions that were conditional on the prior answer.

A number of participants in this survey will be selected to receive a voucher. To help us determine which voucher to send if you are selected, please tell us which you prefer for each

of the following questions. If you are selected, we will randomly choose one of the following questions and send you the voucher you prefer.

Would you prefer [14 oz. vegetarian ground meat (\$8.99 value)] or [14 oz. conventional ground beef (\$6.60 value)]? *[randomized order]*

If they just selected conventional meat: Would you prefer [28 oz. vegetarian ground meat (\$17.98 value)] or [14 oz. conventional ground beef (\$6.60 value)]? *[randomized order]*

If they just selected conventional meat: Would you prefer [42 oz. vegetarian ground meat (\$26.97 value)] or [14 oz. conventional ground beef (\$6.60 value)]? *[randomized order]*

Alternative: Would you prefer [22 oz. vegetarian ground meat (\$14.98 value)] or [14 oz. conventional ground beef (\$6.60 value)]? *[randomized order]*

Alternative: Would you prefer [14 oz. vegetarian ground meat (\$8.99 value)] or [28 oz. conventional ground beef (\$13.20 value)]? *[randomized order]*

If they just selected vegetarian meat: Would you prefer [14 oz. vegetarian ground meat (\$8.99 value)] or [21 oz. conventional ground beef (\$9.90 value)]? *[randomized order]*

Alternative: Would you prefer [14 oz. vegetarian ground meat (\$8.99 value)] or [42 oz. conventional ground beef (\$19.80 value)]? *[randomized order]*

Products in development: which do you prefer?

For the following options, please select which option you prefer.

Would you prefer [14 oz. clean ground meat] or [14 oz. conventional ground beef]? *[randomized order]*

If they just selected conventional meat: Would you prefer [28 oz. clean ground meat] or [14 oz. conventional ground beef]? *[randomized order]*

If they just selected conventional meat: Would you prefer [42 oz. clean ground meat] or [14 oz. conventional ground beef]? *[randomized order]*

Alternative: Would you prefer [22 oz. clean ground meat] or [14 oz. conventional ground beef]? *[randomized order]*

Alternative: Would you prefer [14 oz. clean ground meat] or [28 oz. conventional ground beef]? *[randomized order]*

If they just selected clean meat: Would you prefer [14 oz. clean ground meat] or [21 oz. conventional ground beef]? *[randomized order]*

Alternative: Would you prefer [14 oz. clean ground meat] or [42 oz. conventional ground beef]? [*randomized order*]

Finally, if you are randomly chosen to receive a prize worth around \$20, you have the option to give a portion of your prize instead as a contribution to a charity that helps prevent harm to farm animals. How much of your \$20 prize would you like to give up?

Options: \$0, \$2, \$4, \$6, \$8, \$10, \$12, \$14, \$16, \$18, \$20

Finally, if you are randomly chosen to receive a prize worth around \$20, you have the option to give a portion of your prize instead as a contribution to a charity that helps prevent harm to farm animals. How much of your \$20 prize would you like to give up?

Options: \$0, \$2, \$4, \$6, \$8, \$10, \$12, \$14, \$16, \$18, \$20

How often in the past month have you thought about the novel coronavirus (COVID-19)?

Options: more than 10 times per day, 6-10 times per day, 1-5 times per day, 2-6 times per week, 1 time per week, less than 1 time per week, do not know what it is

C. Mechanical Turk Sample

Prior to the study described in the body of the paper, we conducted two closely related studies on Amazon Mechanical Turk in 2016 and 2017. Below we present the design of the studies and present results from the earlier samples where they can be compared with the more recent study. Both studies were pre-registered with the American Economic Association (AEA) Randomized Controlled Trial (RCT) registry and pre-analysis plans posted at the Open Science Framework (OSF).

C.1. Study 1

The first study randomly exposed some respondents to cultured meat, some to a meat substitute implied to be plant-based, and some to no product but without any additional priming.

C.1.1. Design

In the first study, we recruited a total of 1,800 participants from Amazon’s Mechanical Turk (MTurk) to participate in a two-wave survey. We restricted participation to US-based individuals, since cultured meat products are expected to first be released in the United States and the United States is the world’s top consumer of meat products (OECD, 2017).

All participants completed a baseline survey measuring demographic characteristics, current eating habits, attitudes towards meat products and substitutes, and other relevant variables. Participants were then randomly divided into several treatment arms. One set of participants was exposed to information about cultured meat alongside a persuasive message about the health, environmental, or ethical advantages of reducing consumption of conventional farm animal products. Another set was instead told about a new “animal-free” meat substitute with the same qualities, but they were not explicitly told that it was a cultured meat product and they were told about the product in the context of vegetarian substitutes. Since cultured meat products are new and not many people know about them, respondents not explicitly told about them would be expected to infer the product was a standard vegetarian product. A third set of respondents was provided the same information without mention of a new product. The three “product” treatment arms (cultured meat, vegetarian substitute, no new product) were crossed with the three “message” treatment arms (health, environmental, ethical) with 200 participants in each cell, for a total of 1,800 participants.

Participants were then asked to watch a 90-second video about factory farming which provided more information about various ways in which it can harm farm animals. We then elicited a second round of ethical beliefs, asked some questions about expected future con-

sumption, and asked if participants would like more information about vegetarian products or, if they received information about a new product, if they would like to know when the product that was described became available in their area.

One month later, a follow-up survey was conducted. Respondents were notified that the survey was available by a series of three e-mails, and 84% of those invited to take the survey completed it. The follow-up survey was intended to allow us both to track outcomes over time and to reduce the risk of social desirability bias. Respondents were again asked questions about their current consumption of animal products and about their ethical beliefs.

C.1.2. Results

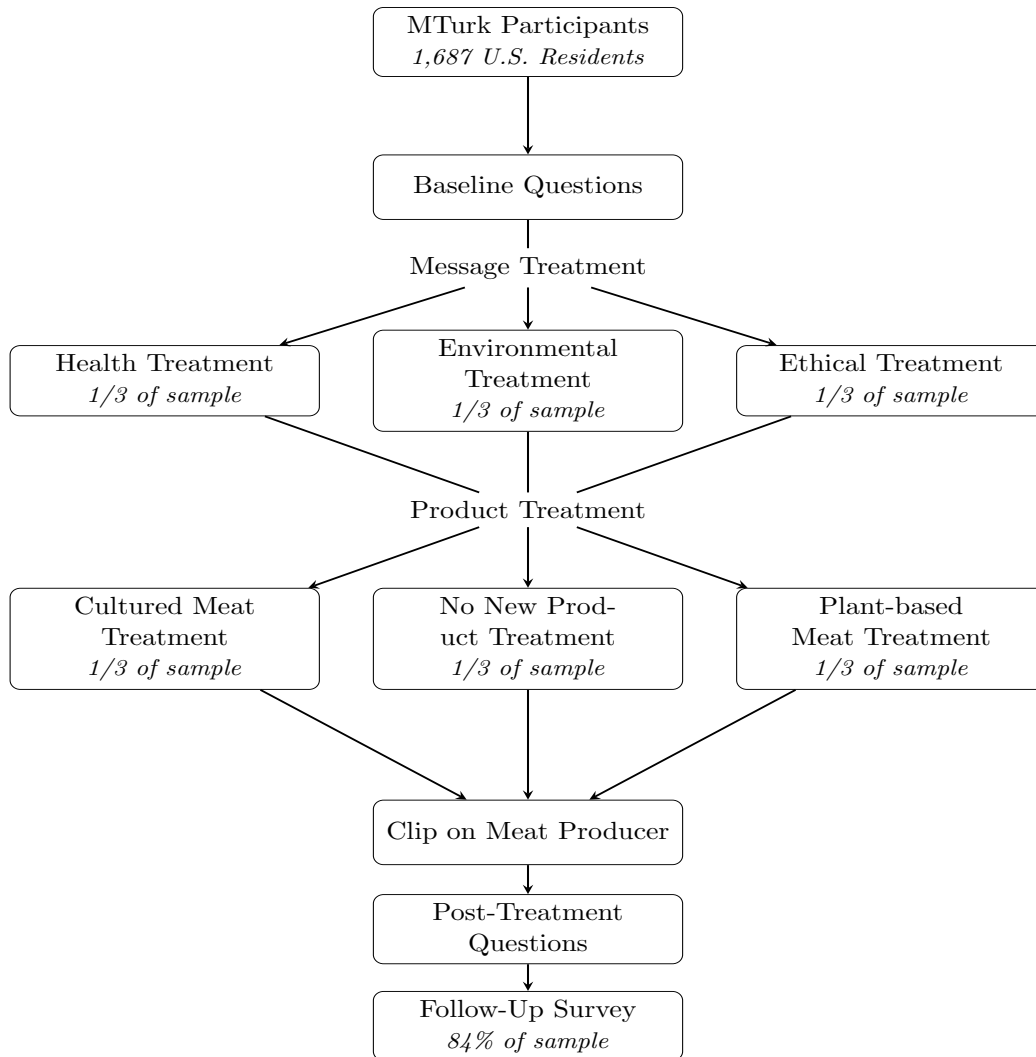
Below, we present the results of the first Mechanical Turk study that can be compared with the results in the body of the paper. In contrast to the results in the main paper, donations were elicited out of a 20-cent bonus rather than out of \$20. We noted that the cultured meat treatment arm had if anything negative impacts on moral concern for animal welfare, however, descriptively this appeared to be driven by the large number of individuals who felt negatively towards the product. This encouraged us to run a second study in which positivity towards the new product was experimentally manipulated.

Table A1: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 1

	(1) Morally preferable	(2) Donation (in cents)
Cultured meat	-0.171** (0.09)	0.100 (0.26)
Plant-based meat	0.065 (0.09)	-0.145 (0.25)
Observations	1687	1687

Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents view plant-based eating as morally preferable and willingness to donate to an animal charity. Column (1) controls for the baseline value of the outcome variable.

Figure A5: MTurk Study 1 Experimental Design



Notes: This figure shows the experimental design for the first MTurk study. The message and product treatments are fully cross-randomized.

Table A2: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 1

Animal-related outcomes		
	(1) Belief about harm	(2) Importance of harm
Cultured meat	0.004 (0.05)	0.005 (0.05)
Plant-based meat	-0.090* (0.05)	0.004 (0.05)
Observations	1687	1687
Environmental outcomes		
	(1) Belief about harm	(2) Importance of harm
Cultured meat	0.003 (0.05)	-0.005 (0.07)
Plant-based meat	-0.058 (0.05)	-0.028 (0.06)
Observations	1687	1687

Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents view animal farming as harming animals and the importance of that harm in the top panel, and analogous questions for the environment in the bottom panel. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption and the baseline value of the outcome variable.

Table A3: Regression of Change in Beliefs on Treatments, MTurk Study 1

	(1) Expects to reduce consumption	(2) How much expects to reduce
Cultured meat	-0.206 (0.20)	-0.272 (0.20)
Plant-based meat	-0.223 (0.20)	-0.222 (0.20)
Observations	1687	1687

Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents expect to reduce their meat consumption and how much they expect to reduce it. Both columns control for a six-tier categorical variable for the respondent's baseline meat consumption variable.

C.2. Study 2

The second study crossed the original treatments with different framings of the substitute and, in a follow-up survey, an anti-cultured meat “unnatural” priming and arguments addressing concerns with naturalness.

C.2.1. Design

Due to the surprising results of the initial study, we conducted another experiment in which we made several changes to the protocol. First, we were concerned that some people might be turned off by the description of the cultured meat product. Thus, in the second iteration we referred to the product as “clean” meat, rather than “cultured” meat as in Study 1, as research by industry groups suggests people have a more favourable response to the product when described this way. Second, we wondered if individuals in the vegetarian substitute arm might be more likely to exhibit more change than those in the cultured meat arm due to social desirability bias or due to ethical issues having more salience, since many people know vegetarians who avoid meat for ethical reasons. To mitigate this concern, we randomly assigned individuals to receive information in one of three framings: an “ethical” framing, a “process” framing, or a “neutral” framing. In the ethical framing arm, we asked respondents to consider whether they knew anyone who was vegetarian for ethical reasons and asked them to reflect on that; those receiving the process framing treatment were the only ones who were still provided additional details about the process of creating cultured meat cut for the other groups; the neutral framing group received neither the ethical questions nor the process description. If a framing or salience explanation were driving results, we might expect to observe differences between these treatment arms. Finally, we focused on an experimental priming treatment which took some of the open-ended text responses from previous participants about their negative feelings towards the product and presented these responses to participants.

This experiment was embedded in another study on the naturalistic heuristic which considered how alternative messages could help to overcome a bias against the “unnatural”

(Macdonald and Vivalt, 2017). We obtained 4,000 responses on MTurk, but due to the focus of this study, 3,200 of these were in the cultured meat treatment arm, with only 400 receiving the vegetarian substitute treatment and another 400 receiving no new product information.

Respondents first answered questions about their ethical beliefs, the material beliefs they held about the harms caused to animals or the environment through conventional farm animal agriculture, and the personal weights they placed on these harms. They were then exposed to a new product treatment arm (or no new product), as before. They also saw the same video and were again asked questions about their beliefs.

Approximately 1 week later, those who had been assigned to one of the cultured meat treatment arms were contacted to participate in a follow-up survey that primed half the participants with negative social information about cultured meat and provided additional information countering the naturalistic heuristic. In particular, participants assigned to receive negative social information were shown five short quotes from previous respondents expressing negative sentiments about cultured meat (e.g. “This seems very unnatural. I don’t feel comfortable about this.”, “Artificial meat sounds disgusting”). Participants who were not assigned to receive negative social information were shown nothing instead. For the messages countering the naturalistic heuristic, participants were asked to read an article of approximately 150-200 words, containing three images that helped to convey the main message. Participants were cross-randomized into receiving one of four messages: a placebo message, a debunking unnatural appeal, an embrace unnatural appeal, and a descriptive norm appeal. These appeals are described elsewhere (Macdonald and Vivalt, 2017).

The alternative appeals to overcome the naturalistic heuristic could have been used as an instrument for the perceived quality of the substitute, but unfortunately these strategies did not have any effect on results. What was a strong instrument for how positively participants viewed cultured meat was simply whether they had been primed with negative social information ($F=24.32$). We thus use this as a lever to consider the relationship between new products and changes in ethical beliefs.

C.3. Results

Below, we present the results of the second Mechanical Turk study that can be compared with the results in the body of the paper. Since the priming only happened in a follow-up survey, we first present results for the effect of cultured meat information on attitudes in the baseline survey and then the effects of priming on attitudes for the follow-up survey, which only included respondents assigned to the cultured meat treatment

C.3.1. Cultured Meat Effects (Baseline Survey)

The first set of results present the effect of cultured meat information on moral attitudes and finds effects that are broadly similar to those in the main paper. Again, donations in the MTurk experiment were elicited out of a 20-cent bonus rather than out of \$20.

Table A4: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 2

	(1) Morally preferable	(2) Donation (in cents)
Cultured meat	-0.075 (0.07)	-1.068** (0.50)
Plant-based meat	0.011 (0.10)	-1.035 (0.66)
Observations	3234	3233

Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents view plant-based eating as morally preferable and willingness to donate to an animal charity. Column (1) controls for the baseline value of the outcome variable.

Table A5: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 2

Animal-related outcomes		
	(1) Belief about harm	(2) Importance of harm
Cultured meat	-0.068 (0.05)	-0.007 (0.05)
Plant-based meat	-0.042 (0.06)	0.051 (0.07)
Observations	3233	3233
Environmental outcomes		
	(1) Belief about harm	(2) Importance of harm
Cultured meat	-0.036 (0.05)	-0.098* (0.05)
Plant-based meat	-0.008 (0.07)	-0.059 (0.07)
Observations	3232	3232

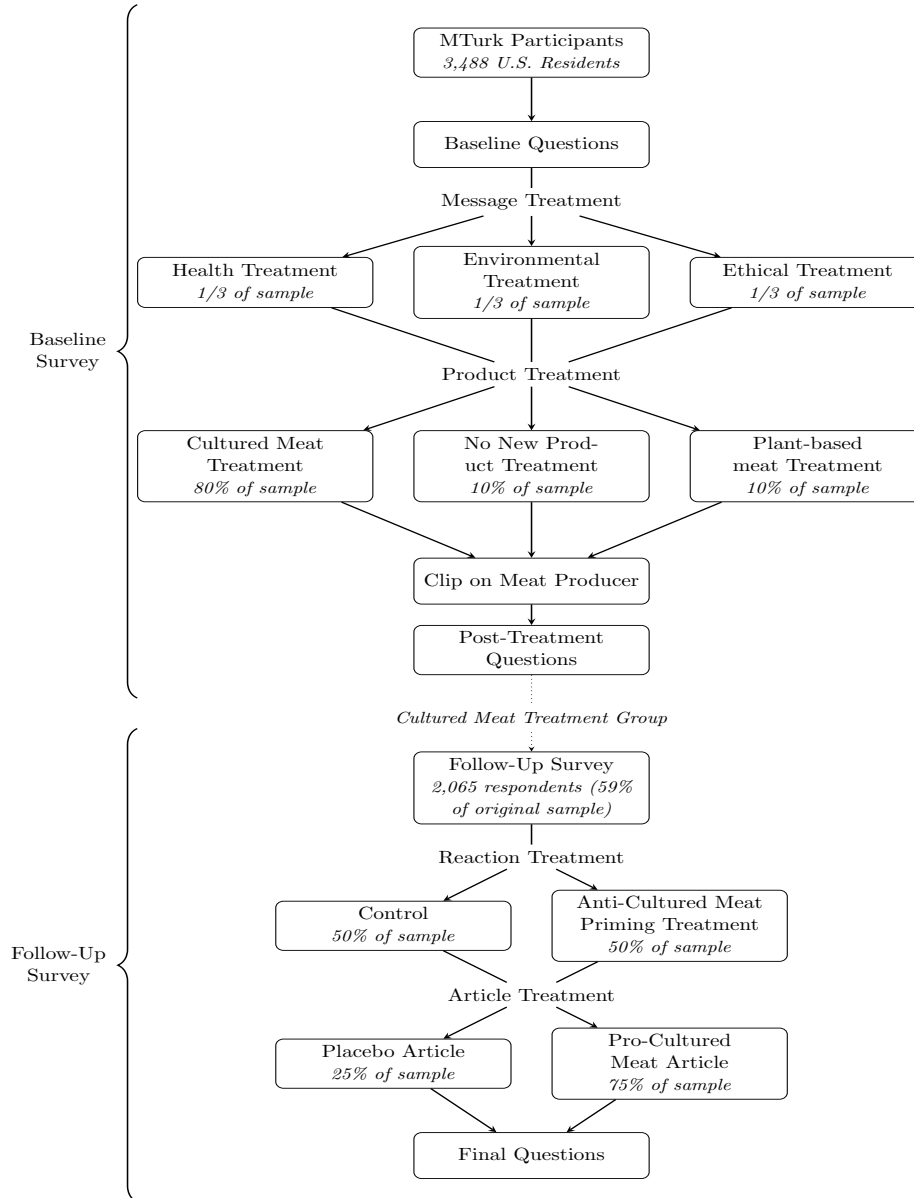
Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents view animal farming as harming animals and the importance of that harm in the top panel, and analogous questions for the environment in the bottom panel. Both columns control for a six-tier categorical variable for the respondent's baseline meat consumption and the baseline value of the outcome variable.

Table A6: Regression of Change in Beliefs on Treatments, MTurk Study 2

	(1) Expects to reduce consumption	(2) How much expects to reduce
Cultured meat	-0.511*** (0.20)	-0.472** (0.19)
Plant-based meat	-0.394 (0.26)	-0.284 (0.25)
Observations	3225	3224

Notes: The table shows results from a prior study on Amazon Mechanical Turk. The columns show the effect of assignment to information about cultured meat and information about plant-based meat on whether respondents expect to reduce their meat consumption and how much they expect to reduce it. Both columns control for a six-tier categorical variable for the respondent's baseline meat consumption variable.

Figure A6: MTurk Study 2 Experimental Design



Notes: The pro-cultured meat article is one of three: an article arguing that naturalness is meaningless and not necessarily good (“debunk unnaturalness”), an article arguing that many good and popular things are unnatural (“embrace unnaturalness”), or an article claiming that many people are interested in cultured meat. The treatments are considered in a companion paper but none has any substantive effects and they would make a poor instrument for how positively a participant feels about the cultured meat product (Macdonald and Vivaldi 2017). The placebo article is an article on the benefits of exercise. Within the cultured meat arm, the priming treatment (priming vs. no priming) and article treatment (placebo vs. pro-cultured meat) are fully cross-randomized.

C.3.2. Priming Effects

The second set of results present the effect of cultured meat information on moral attitudes and finds effects that are broadly similar to those in the main paper.

Table A7: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 2

	(1) Morally preferable OLS	(2) Morally preferable OLS	(3) Morally preferable 2SLS	(4) Donation (in cents) OLS	(5) Donation (in cents) OLS	(6) Donation (in cents) 2SLS
Anti-Cultured Meat Priming	-0.161*** (0.06)			0.851** (0.38)		
Pro-Cultured Meat Article	-0.128* (0.07)			0.505 (0.45)		
Change in Positivity toward Cultured Meat		0.035** (0.01)	0.143*** (0.05)		0.301*** (0.08)	-0.767** (0.34)
Positivity toward Cultured Meat (Pre-Video)		-0.006 (0.01)	0.022 (0.01)		0.417*** (0.05)	0.155 (0.09)
Observations	2065	2065	2065	2065	2065	2065

Notes: The table shows results from a prior study on Amazon Mechanical Turk. Columns (1) and (4) show the effect of assignment to anti-cultured meat social priming and an pro-cultured meat article, and information about plant-based meat on whether respondents view plant-based eating as morally preferable and willingness to donate to an animal charity, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it. Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption, and Columns (1)-(3) control for the baseline value of the outcome variable.

Table A8: Regression of Change in Beliefs and Donations on Treatments, MTurk Study 2

Animal-related outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Anti-Cultured Meat Priming	-0.044 (0.04)			-0.075** (0.03)		
Pro-Cultured Meat Article	0.006 (0.04)			-0.004 (0.04)		
Change in Positivity toward Cultured Meat		0.012 (0.01)	0.037 (0.03)		0.020** (0.01)	0.065** (0.03)
Positivity toward Cultured Meat (Pre-Video)		0.011** (0.00)	0.017** (0.01)		-0.000 (0.00)	0.011 (0.01)
Observations	2065	2065	2065	2065	2065	2065
Environmental outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Anti-Cultured Meat Priming	-0.033 (0.04)			-0.028 (0.04)		
Pro-Cultured Meat Article	0.117*** (0.04)			0.092** (0.04)		
Change in Positivity toward Cultured Meat		0.014* (0.01)	0.024 (0.03)		0.011 (0.01)	0.020 (0.03)
Positivity toward Cultured Meat (Pre-Video)		0.009* (0.00)	0.012 (0.01)		0.016*** (0.00)	0.018* (0.01)
Observations	2065	2065	2065	2065	2065	2065

Notes: The table shows results from a prior study on Amazon Mechanical Turk. Columns (1) and (4) show the effect of assignment to information about anti-cultured meat social priming and an pro-cultured meat article on whether respondents view animal farming as harming animals and the importance of that harm in the top panel, and analogous questions for the environment in the bottom panel, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it. Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption and the baseline value of the outcome variable.

Table A9: Regression of Change in Beliefs on Treatments, MTurk Study 2

	(1) Expects to reduce consumption OLS	(2) Expects to reduce consumption OLS	(3) Expects to reduce consumption 2SLS	(4) How much expects to reduce OLS	(5) How much expects to reduce OLS	(6) How much expects to reduce 2SLS
Anti-Cultured Meat Priming	-0.124 (0.14)			-0.075 (0.13)		
Pro-Cultured Meat Article	0.269* (0.16)			0.237 (0.15)		
Change in Positivity toward Cultured Meat		0.159*** (0.03)	0.088 (0.11)		0.136*** (0.03)	0.048 (0.11)
Positivity toward Cultured Meat (Pre-Video)		0.183*** (0.02)	0.165*** (0.03)		0.169*** (0.02)	0.147*** (0.03)
Observations	2065	2065	2065	2065	2065	2065

Notes: The table shows results from a prior study on Amazon Mechanical Turk. Columns (1) and (4) show the effect of assignment to information about anti-cultured meat social priming and an pro-cultured meat article on whether respondents expect to reduce their meat consumption and how much they expect to reduce it, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it. Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption.

Table A10: First Stage: Regression of Positivity and Easiness Indices on No Negative Priming Treatment, MTurk Study 2

	(1) Change in Positivity toward Cultured Meat	(2) Easiness index
No Anti-Cultured Meat Priming	1.168*** (0.10)	0.055 (0.14)
Positivity toward Cultured Meat (Pre-Video)	-0.247*** (0.01)	0.173*** (0.02)
Observations	2065	2065

Notes: The table presents results from a prior study on Amazon Mechanical Turk. The table shows the effect of not receiving anti-cultured meat social priming on how positively respondents view plant-based eating and how easy they believe it would be to reduce their meat consumption. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Easiness is measured by the sum of two seven-point Likert scale questions assessing how easy respondents think it would be to eliminate farm meat from their diet and to reduce it (after flipping the scale so that higher is better).

D. One-Month Follow-up

One month after the initial survey, we circulate a follow-up survey to all respondents. The follow-up survey reminds them of the product, asks them about their familiarity with it, and then asks the same questions about attitudes, beliefs, (incentivized) willingness to donate to an animal charity, and (incentivized) willingness to pay for farm meat compared to plant-based or cultured meat. Below are results from the follow up for comparison with those in the main paper.

Table A11: First Stage: Regression of Positivity and Easiness Indices on No Negative Priming Treatment (Follow-Up)

	(1)	(2)
	Change in Positivity toward Cultured Meat	Easiness index
No Anti-Cultured Meat Priming	0.244*** (0.05)	0.144 (0.10)
Positivity toward Cultured Meat (Pre-Video)	-0.387*** (0.01)	0.402*** (0.02)
Observations	3886	3886

Notes: The table presents results from a follow-up survey one month after the baseline. The table shows the effect of not receiving anti-cultured meat social priming on how positively respondents view plant-based eating and how easy they believe it would be to reduce their meat consumption. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Easiness is measured by the sum of two seven-point Likert scale questions assessing how easy respondents think it would be to eliminate farm meat from their diet and to reduce it (after flipping the scale so that higher is better).

Table A12: Regression of Change in Beliefs and Donations on Treatments (Follow-Up)

	(1) Morally preferable OLS	(2) Morally preferable OLS	(3) Morally preferable 2SLS	(4) Donation (in dollars) OLS	(5) Donation (in dollars) OLS	(6) Donation (in dollars) 2SLS
Cultured Meat	-0.132* (0.07)			-0.176 (0.48)		
Anti-Cultured Meat	-0.057 (0.04)			0.155 (0.26)		
Priming	0.074 (0.05)			0.132 (0.30)		
Pro-Cultured Meat	0.055 (0.08)			-0.225 (0.53)		
Plant-based Meat						
Change in Positivity toward Cultured Meat		0.178*** (0.01)	0.213 (0.16)		0.395*** (0.08)	-0.385 (1.08)
Positivity toward Cultured Meat (Pre-Video)		0.219*** (0.01)	0.237*** (0.07)		0.804*** (0.06)	0.502 (0.42)
Observations	4846	3886	3886	4845	3885	3885

Notes: The table presents results from a follow-up survey one month after the baseline. Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents view plant-based eating as morally preferable and willingness to donate to an animal charity out of a \$20 bonus, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption, and Columns (1)-(3) control for the baseline value of the outcome variable.

Table A13: Regression of Change in Beliefs and Donations on Treatments (Follow-Up)

Animal-related outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Cultured Meat	-0.042 (0.06)			0.081 (0.05)		
Anti-Cultured Meat	-0.027 (0.03)			0.006 (0.03)		
Priming						
Pro-Cultured Meat	-0.010 (0.04)			-0.064* (0.04)		
Article						
Plant-based Meat	-0.062 (0.06)			0.043 (0.06)		
Change in Positivity toward Cultured Meat		0.099*** (0.01)	0.131 (0.12)		0.125*** (0.01)	0.004 (0.12)
Positivity toward Cultured Meat (Pre-Video)		0.132*** (0.01)	0.144*** (0.05)		0.128*** (0.01)	0.077 (0.05)
Observations	4846	3886	3886	4846	3886	3886
Environmental outcomes						
	(1) Belief about harm OLS	(2) Belief about harm OLS	(3) Belief about harm 2SLS	(4) Importance of harm OLS	(5) Importance of harm OLS	(6) Importance of harm 2SLS
Cultured Meat	0.067 (0.06)			0.034 (0.06)		
Anti-Cultured Meat	0.008 (0.03)			0.010 (0.03)		
Priming						
Pro-Cultured Meat	-0.043 (0.04)			-0.050 (0.04)		
Article						
Plant-based Meat	-0.030 (0.06)			-0.036 (0.06)		
Change in Positivity toward Cultured Meat		0.111*** (0.01)	-0.007 (0.12)		0.126*** (0.01)	-0.010 (0.13)
Positivity toward Cultured Meat (Pre-Video)		0.139*** (0.01)	0.088* (0.05)		0.151*** (0.01)	0.092* (0.05)
Observations	4845	3886	3886	4846	3886	3886

Notes: The table presents results from a follow-up survey one month after the baseline. Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents view animal farming as harming animals and the importance of that harm in the top panel, and analogous questions for the environment in the bottom panel, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption and the baseline value of the outcome variable.

Table A14: Regression of Change in Beliefs on Treatments (Follow-Up)

	(1) Expects to reduce consumption OLS	(2) Expects to reduce consumption OLS	(3) Expects to reduce consumption 2SLS	(4) How much expects to reduce OLS	(5) How much expects to reduce OLS	(6) How much expects to reduce 2SLS
Cultured Meat	-0.402** (0.19)			-0.300 (0.18)		
Anti-Cultured Meat Priming	-0.092 (0.11)			-0.069 (0.10)		
Pro-Cultured Meat Article	0.084 (0.12)			-0.000 (0.12)		
Plant-based Meat	-0.242 (0.21)			-0.203 (0.20)		
Change in Positivity toward Cultured Meat		0.607*** (0.03)	0.569 (0.37)		0.583*** (0.03)	0.449 (0.36)
Positivity toward Cultured Meat (Pre-Video)		0.729*** (0.02)	0.715*** (0.15)		0.657*** (0.02)	0.605*** (0.14)
Observations	4846	3886	3886	4846	3886	3886

Notes: The table presents results from a follow-up survey one month after the baseline. Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on whether respondents expect to reduce their meat consumption and how much they expect to reduce it. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption.

Table A15: Regression of Relative WTP for Farm vs. Alternative Meat on Treatments (Follow-Up)

	(1) Log WTP (vs. plant-based) OLS	(2) Log WTP (vs. plant-based) OLS	(3) Log WTP (vs. plant-based) 2SLS	(4) Log WTP (vs. cultured) OLS	(5) Log WTP (vs. cultured) OLS	(6) Log WTP (vs. cultured) 2SLS
Cultured Meat	0.119* (0.07)					
Anti-Cultured Meat	0.017 (0.04)			0.091** (0.04)		
Priming						
Pro-Cultured Meat	-0.096** (0.04)			-0.023 (0.05)		
Article						
Plant-based Meat	-0.009 (0.08)					
Change in Positivity toward Cultured Meat		-0.168*** (0.01)	-0.140 (0.14)		-0.252*** (0.01)	-0.465*** (0.14)
Positivity toward Cultured Meat (Pre-Video)		-0.248*** (0.01)	-0.237*** (0.06)		-0.323*** (0.01)	-0.406*** (0.05)
Observations	4846	3886	3886	3885	3885	3885

Notes: The table presents results from a follow-up survey one month after the baseline. Columns (1) and (4) show the effect of assignment to information about cultured meat, anti-cultured meat social priming, an pro-cultured meat article, and information about plant-based meat on the logarithm of respondents' willingness to pay per unit of plant-based meat and cultured meat, respectively, in units of farm meat, respectively. Columns (2) and (5) show the degree to which respondents' baseline positivity about cultured meat and change in positivity between the start and end of the survey predict the respective outcomes. Positivity toward cultured meat is measured by the sum of two five-point Likert scale questions assessing respondents' interest in cultured meat and whether they say they would eat it (after flipping the scale so that higher is better). Columns (3) and (6) instrument for the change in positivity using the anti-cultured meat priming. All columns control for a six-tier categorical variable for the respondent's baseline meat consumption.

E. Additional Figures and Tables

Table A16: Raw Means of Outcome Variables, by Cultured Meat Treatment Articles

	Control article	Embrace Unnatural article	p-value (vs. control article)	Debunk Unnatural article	p-value (vs. control article)	Descriptive Norm article	p-value (vs. control article)
Pre-treatment							
Morally preferable (1-7 scale) ^R	4.116	4.132	0.822	4.159	0.554	4.112	0.957
Belief about harm (1-5 scale) ^R	2.884	2.818	0.197	2.865	0.706	2.808	0.140
Harm is important (1-5 scale) ^R	3.284	3.281	0.964	3.295	0.823	3.246	0.466
Post-treatment							
Morally preferable (1-7 scale) ^R	3.617	3.636	0.813	3.612	0.951	3.568	0.525
Belief about harm (1-5 scale) ^R	2.212	2.180	0.526	2.158	0.282	2.177	0.491
Harm is important (1-5 scale) ^R	2.551	2.525	0.632	2.539	0.822	2.543	0.880
Expects to reduce (0-10 scale)	5.243	5.228	0.911	5.074	0.230	5.134	0.434
How much reduce (0-10 scale)	4.861	4.867	0.961	4.760	0.468	4.758	0.457
Donation (dollars)	9.408	9.568	0.639	9.305	0.764	9.902	0.148
Log WTP (for farm vs. plant-based meat) ^R	0.175	0.126	0.331	0.096	0.126	0.101	0.146
Log WTP (for farm vs. cultured meat) ^R	-0.063	-0.085	0.678	-0.078	0.767	-0.134	0.174

Notes: This table presents mean values, pre- and post-treatment, for the main outcome variables, by the articles treatment, restricting attention to those in the cultured meat treatment arm. The pre-treatment values are prior to assignment to the farm meat, plant-based meat, or cultured meat treatment arms (before assignment to different articles), while the post-treatment values are from questions asked following the video. The maximum and minimum values of the Likert scales are as follows: for whether eating vegetarian is “Morally preferable”, 1 = strongly agree, 7 = strongly disagree; for whether animals are harmed in conventional meat production (“Belief about harm”), 1 = a great deal, 5 = not at all; for whether this harm is important (“Harm is important”), 1 = a great deal, 5 = not at all; for whether the participant “expects to reduce” their meat consumption, “0 = definitely not, 10 = definitely yes”; for how much they expect to reduce it (“How much reduce”), 0 = not at all, 10 = a lot. The full Likert scales are as in the description of the YouGov experiment in the appendix. Raw values are presented, but those outcome variables for which a high value represents low concern for animals or stronger preferences towards farm meat have a superscript ^R next to them indicating this. Whether participants expected to reduce their consumption of farm meat and by how much, as well as their donation and WTP for farm meat vs. plant-based meat, was only elicited post-treatment.